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✉ For correspondence:

ishii-k@g.ecc.u-tokyo.ac.jp

emoto@bs.s.u-tokyo.ac.jp

* Lead contact

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Neuropeptidergic circuit modulation of developmental sleep in *Drosophila*

Chikayo Hemmi¹, Kenichi Ishii¹✉, Mana Motoyoshi¹, Masato Tsuji¹, Kazuo Emoto^{1,2,*}✉

¹Department of Biological Sciences, The University of Tokyo, Tokyo, Japan • ²International Research Center for Neurointelligence (WPI-IRCN), The University of Tokyo, Tokyo, Japan

eLife Assessment

The study investigates an emerging research field: the interaction between sleep and development. The authors used *Drosophila* larvae sleep as a study model and provide insight into how neuropeptide circuitry control sleep differentially between larvae and adult *Drosophila*. By using board range of behaviour and imaging methods and analysis, the authors provide a **valuable** investigation that demonstrates a larvae-specific sleep regulatory neural pathway of Hugin-PK2-Dilps in the *Drosophila* neurosecretory centre IPC. While some further text clarifications are still required, the revision presented **convincing** evidence supporting the claims with the new imaging data, sleep parametric analysis, and further clarification addressing the reviewers' comments.

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Abstract

Sleep-wakefulness regulation dynamically evolves along development in a wide range of organisms. While the mechanism regulating sleep in adults are relatively well understood, little is known about its counterpart in early developmental stages. Here we report a neuropeptidergic circuitry that modulates sleep in developing *Drosophila* larvae. Through an unbiased screen, we identified the neuropeptide Hugin and its receptor PK2-R1 as critical regulators of larval sleep. Our genetic and behavioral data suggest that HugPC neurons secrete Hugin peptides to activate Insulin-producing cells (IPCs), which expresses a Hugin receptor PK2-R1. IPCs, in turn, release insulin-like peptides (Dilps) to regulate sleep. We further show that the Hugin/PK2-R1 axis is dispensable for adult sleep. Our findings thus reveal the neuromodulatory circuit that regulates developmental sleep in larvae, and highlight differential impacts of the same modulatory axis on early-life sleep and adult sleep.

Introduction

Sleep is a fundamental state with significant impacts on many aspects of cognition and metabolism (Blumberg et al., 2005 [↗](#); Davis et al., 2004 [↗](#); Sorribes et al., 2013 [↗](#)). While sleep remains essential throughout life, its regulation dynamically unfolds over the course of development. For instance, the pace of sleep-wakefulness cycle is relatively fast in infants, while gradually slows down and stabilize in adults (Blumberg et al., 2005 [↗](#); Davis et al., 2004 [↗](#); Sorribes et al., 2013 [↗](#)). Furthermore, sleep-wake cycle in infants is typically independent of circadian rhythm, but it gradually comes under the circadian regulation (Davis et al., 2004 [↗](#); Frank et al., 2017 [↗](#); Poe et al., 2023 [↗](#)). Such developmental evolution of sleep-wakefulness cycles appears to be conserved across a wide range of organisms including flies, fishes, mammals and humans (Blumberg et al., 2005 [↗](#); Davis et al., 2004 [↗](#); Poe et al., 2023 [↗](#); Sorribes et al., 2013 [↗](#); Szuperak et al., 2018 [↗](#)). In contrast to deep insights into the molecular and neural mechanisms underlying adult sleep regulation (Barlow & Rihel, 2017 [↗](#); Scammell et al., 2017 [↗](#); Shafer & Keene, 2021 [↗](#)), little is known about regulatory mechanisms underlying developmental sleep in infants, partially due to a lack of convenient models to study developmental sleep.

Drosophila larvae have recently emerged as a suitable model to study sleep regulation mechanisms. Recent studies have reported that the second instar larvae show short periods (< 6 seconds) of “inactive state” during locomotion. Importantly, such “inactive state” is consistent with the general definition of sleep: reduced responsiveness to noxious stimuli, homeostatic response to sleep deprivation, and rapid reversibility upon stimulation (Szuperak et al., 2018 [DOI](#)), suggesting that the “inactive state” in larvae likely corresponds to the sleep state. Interestingly, loss-of-function mutations in the clock genes *clock* and *cyc* fail to impact the larval sleep, while significantly altering the sleep patterns in adult flies (Dubowy & Sehgal, 2017 [DOI](#); Hendricks et al., 2003 [DOI](#); Szuperak et al., 2018 [DOI](#)). Likewise, mutations in the dopamine transporter *DAT* fail to influence larval sleep, while reducing sleep amounts in adult (Kume et al., 2005 [DOI](#); Szuperak et al., 2018 [DOI](#)). These observations imply that sleep regulation mechanisms might be at least in part distinct between larvae and adults, yet the neural mechanism of larval sleep remains vastly understudied compared to that of adult sleep.

In this study, we performed an unbiased genetic screen in the second instar larvae, and identified the neuropeptide Hugin and its receptor PK2-R1 as a pair critical for larval sleep. At the circuit level, Hugin-producing HugPC neurons directly stimulate the insulin-producing cells (IPCs) via PK2-R1 to regulate sleep. We further found that regulators of larval sleep are either irrelevant or exert the opposite effects on adult sleep. Overall, our findings uncover the neuropeptidergic circuitry that regulates developmental sleep in larvae, and highlight mechanistic differences between larval and adult sleep.

Results

PK2-R1 is required for wake/sleep control in *Drosophila* larvae

To understand molecular and neural mechanisms of larval sleep, we modified a previously reported system (Churgin et al., 2019 [DOI](#); Szuperak et al., 2018 [DOI](#)) to automatically quantify the sleep amount of second instar larvae (Figure 1-figure supplement 1A-1C [DOI](#)) (see Materials & Methods for details). Our system automatically annotates larvae in each frame as either “inactive” or “active” states on the basis of pixel changes, with ~90% accuracy compared to manual annotation (Figure 1-figure supplement 1C [DOI](#); see Methods for details). We found that bouts of ≥ 12 inactive frames are consistent with the general criteria for sleep; reduced responsiveness to noxious stimuli, homeostatic response to sleep loss, and rapid reversibility upon stimulation (Raizen et al., 2008 [DOI](#); Szuperak et al., 2018 [DOI](#)) (Figure 1-figure supplement 2B-D [DOI](#)). Therefore, the present study defines sleep as inactive state of ≥ 12 consecutive frames (Figure 1-figure supplement 1A [DOI](#)).

Using this system, we next carried out an unbiased screen for genes involved in sleep regulation. To this end, we tested 47 CRISPR knock-out lines of genes encoding enzymes or receptors for neuropeptides and monoamines (Deng et al., 2019 [DOI](#)) (Supplemental Table 1). As illustrated in Figure 1-figure supplement 3 [DOI](#), 13 out of 47 homozygous null mutants exhibited significant changes in sleep amounts compared to the control group. To further test the functions of these candidate genes, we next took advantage of the CRISPR knock-in drivers in which GAL4 sequence is inserted into each gene locus. With these drivers, we silenced neurons expressing each candidate gene by expressing the inward-rectifying potassium channel Kir2.1 (Baines et al., 2001 [DOI](#)). Silencing neurons expressing *Oamb* and *PK2-R1* resulted in significant sleep increase, phenocopying the knock-out of these genes (Figure 1A [DOI](#) and 1B [DOI](#) and Figure 1-figure supplement 4D [DOI](#)). *Oamb* encodes the octopamine receptor and has been reported to be involved in larval sleep (Szuperak et al., 2018 [DOI](#)). The role of PK2-R1 in larval sleep, on the other hand, has been unknown to date. Given its strong expression in insulin-producing cells (Schlegel et al., 2016 [DOI](#)) and its function as a receptor for the neuropeptide Hugin, which modulates feeding (Schoofs et al., 2014 [DOI](#)), we hypothesized that PK2-R1 might mediate neuropeptidergic signaling that links metabolic and sleep regulation during development. We thus focused on this gene as a candidate connecting behavioral and endocrine sleep control.

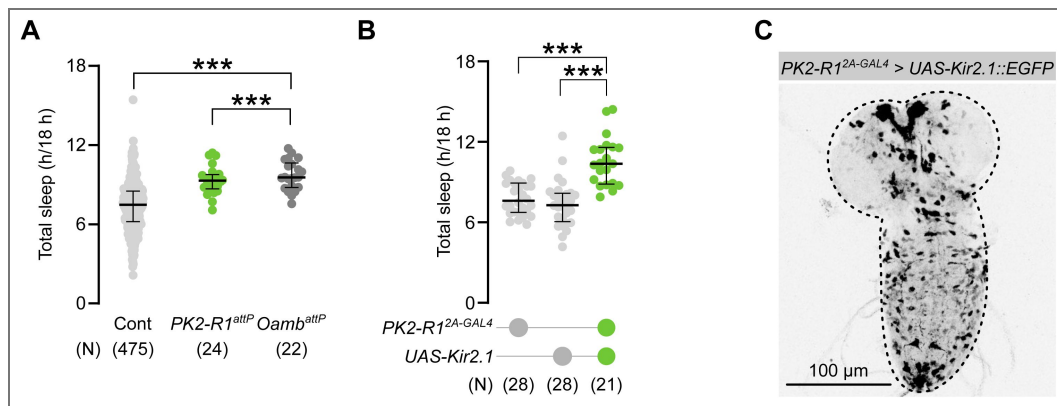


Figure 1. PK2-R1 is required for larval sleep control

(A) Sleep amounts in *PK2-R1* or *Oamb* knockout mutants. Each dot represents an individual animal; in this and the following panels, "N" indicates the number of biologically independent animals per group, and the thick line and thin error bars indicate the median and interquartile range (Q1–Q3), respectively. *** $p < 0.001$ (Mann–Whitney U-test with Bonferroni correction). (B) Sleep amounts in larvae in which *PK2-R1* neurons were silenced. *** $p < 0.001$ (Mann–Whitney U-test with Bonferroni correction). (C) Expression pattern of *PK2-R1-GAL4* > *UAS-Kir2.1::EGFP* larvae.

Insulin-producing cells express PK2-R1 and regulate sleep states in developing larvae

Since the *PK2-R1* driver broadly labeled neurons (~1000 cells) throughout the CNS (Figure 1C [↗](#)), we next attempted to identify a subpopulation of these neurons relevant for sleep regulation. To narrow down the candidates within PK2-R1-positive populations, we examined subsets labeling GABAergic, cholinergic, or glutamatergic neurons, as well as those co-expressing other neuropeptides identified in our screen, but none of these manipulations reproduced the sleep phenotype caused by PK2-R1 perturbation. Notably, insulin-producing cells (IPCs) in the pars intercerebralis (PI), the neurosecretory center of *Drosophila*, have been reported to express *PK2-R1* (Schlegel et al., 2016 [↗](#)). Consistently, the *PK2-R1*^{2A-LexA} knock-in driver labeled all IPCs (Figure 2A and B [↗](#)). To assess whether IPCs are involved in larval sleep, we silenced IPCs with Kir2.1 expression, using the IPC-specific drivers *Dilp3*- and *Dilp5-GAL4* (Figure 2C [↗](#)). We found that silencing of IPCs significantly increased the sleep amounts compared to the control, phenocopying *PK2-R1* neuron silencing and *PK2-R1* knockout (Figure 1A [↗](#)). Consistently, both *Dilp3* and *Dilp5* null mutants exhibited larger sleep amounts compared to the control (Figure 2D [↗](#)). These phenotypes of *Dilp* mutants and IPC silencing are unlikely to be accounted for by locomotion defects, as the travel distances during wake periods were unaffected (Fig. 2—figure supplement 1 [↗](#)). Through these trials, insulin-producing cells (IPCs) emerged as one of the populations whose silencing phenocopied PK2-R1 loss, supporting their key role in larval sleep regulation while not excluding contributions from other PK2-R1-expressing neurons. Collectively, these findings suggest that *PK2-R1* in IPCs regulates larval sleep.

Hugin-expressing neurons control larval sleep states

Given that PK2-R1 is a receptor of the neuropeptide Hugin (Rosenkilde et al., 2003 [↗](#)), we next examined whether Hugin is involved in the larval sleep. We found that *Hug* mutant larvae exhibited significantly larger sleep amounts compared to the control (Figure 3A [↗](#)), consistent with *PK2-R1* knockouts (Figure 1A [↗](#)). When we silenced Hug-expressing neurons by driving the expression of Kir2.1 with the *Hug* CRISPR-knock-in driver *Hug*^{2A-GAL4} (Deng et al., 2019 [↗](#)), it resulted in a significant increase of sleep amounts without detectable changes in locomotion speed (Figure 3B [↗](#) and Figure 3—figure supplement 1B [↗](#)). Conversely, activation of *Hug*-expressing neurons with the red-shifted channel rhodopsin ReaChR (Inagaki et al., 2014 [↗](#)) or heat-sensitive ion channel TrpA1 (Hamada et al., 2008 [↗](#)) both caused a significant reduction of sleep amount (Figure 3C and D [↗](#)) and increased locomotor activity. However, locomotion changes were not consistently observed upon either activation or suppression of Hug neurons (Figure 3—figure supplement 1 [↗](#)), suggesting that changes in sleep cannot be simply explained by locomotor alterations. We further quantified larval feeding using a dye-based ingestion assay and found that silencing HugPC neurons reduced food intake (Figure 3—figure supplement 3 [↗](#)), indicating that the sleep phenotype is unlikely to be explained by feeding behavior being misclassified as sleep. These data together indicate that *Hug*-expressing neurons are responsible for downregulating larval sleep.

Hug-expressing neurons are comprised of 20 cells in the subesophageal zone (SEZ) of the brain, and are classified into four subpopulations based on their projection patterns (Bader et al., 2007 [↗](#)). Of these, one subpopulation called HugPC neurons (Bader et al., 2007 [↗](#)) have been reported to form synaptic connections with IPCs (Hückesfeld, Peters, & Pankratz, 2016 [↗](#)). We thus suspected that HugPC neurons may be responsible for sleep regulation. In line with this notion, axons of HugPC neurons, labeled by a newly generated *HugPC-LexA* driver (Figure 3—figure supplement 2 [↗](#)), and IPCs, labeled by *Dilp3-GAL4* (Figure 3E [↗](#)), exist in space close to each other. Indeed, silencing of HugPC neurons by Kir2.1 significantly increased larval sleep (Figure 3F [↗](#)), phenocopying those observed in *Hugin* and *PK2-R1* mutants. These results hint at the possibility that HugPC neurons activate IPCs via Hugin/PK2-R1 signaling to regulate sleep.

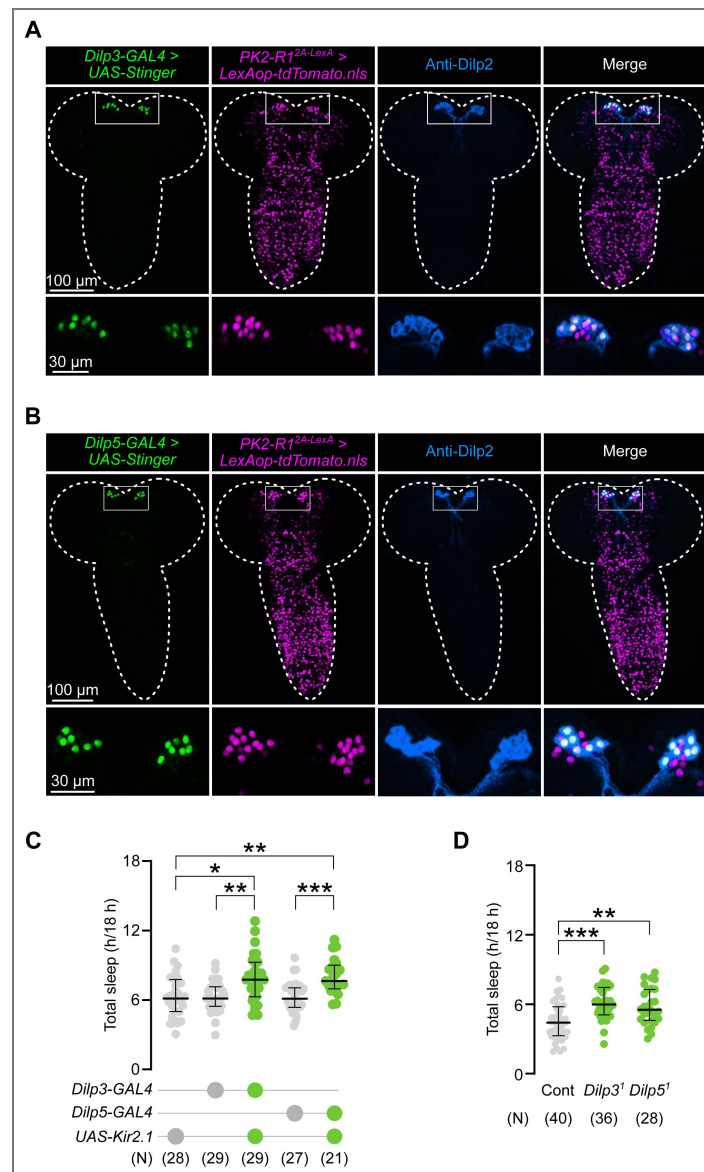


Figure 2. IPCs and Dilps negatively regulate larval sleep

(A) Triple labeling of *PK2-R1* neurons expressing nuclear-localized RFP (magenta), *Dilp3* neurons expressing nuclear-localized GFP (green), and anti-Dilp2 positive cells (blue). Top panels show signals in the larval brain and the VNC. Bottom panels show magnified images of the white-squared area in each top panel, focusing on the dorsomedial brain region where the cell bodies of IPCs are located. Note that all IPCs labeled with *Dilp3-GAL4* overlapped with *PK2-R1^{2A-LexA}*-expressing cells. Similar results were obtained from 5 independent samples of the same genotype. (B) Simultaneous detection of *PK2-R1* neurons expressing nuclear-localized RFP (magenta), *Dilp5* neurons expressing nuclear-localized GFP (green), and anti-Dilp2 positive cells (blue). Similar results were obtained from 5 independent samples of the same genotype. (C) Effect of IPCs silencing on larval sleep. Each dot represents an individual animal; in this and the following panels, "N" indicates the number of biologically independent animals per group, and the thick line and thin error bars indicate the median and interquartile range (Q1-Q3), respectively. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$ (Mann-Whitney U-test with Bonferroni correction). (D) Sleep amounts in *Dilp3* or *Dilp5* null mutants. *** $p < 0.001$, ** $p < 0.01$ (Mann-Whitney U-test with Bonferroni correction).

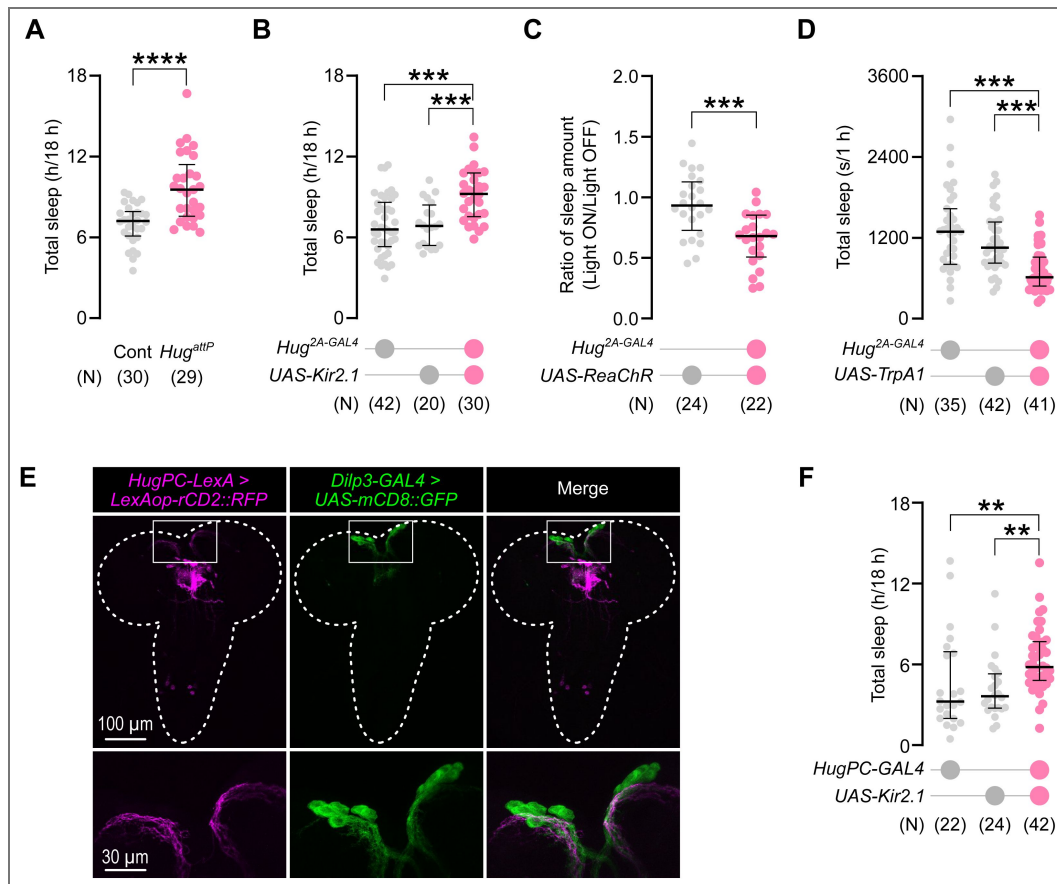


Figure 3. Hug neurons negatively regulate larval sleep

(A) Sleep amounts in *Hug* CRISPR-knockout mutant larvae. **** p < 0.0001 (Mann-Whitney U-test). Each dot represents an individual animal; in this and the following panels, "N" indicates the number of biologically independent animals per group, and the thick line and thin error bars indicate the median and interquartile range (Q1–Q3), respectively. (B) Effect of silencing Hug neurons on larval sleep amount. ** p < 0.01, * p < 0.05 (Mann-Whitney U-test with Bonferroni correction). (C) Larval sleep changes induced by optogenetic activation of Hug neurons. For each genotype, sleep duration in the 1-h light-ON period was normalized to that in the 1-h light-OFF phase. * p < 0.05 (Mann-Whitney U-test with Bonferroni correction). (D) Larval sleep during thermogenetic activation of Hug neurons. ** p < 0.01, * p < 0.05 (Mann-Whitney U-test with Bonferroni correction). (E) Visualization of HugPC neurons and IPCs labeled by rCD2::RFP (magenta) and mCD8::GFP (green), respectively. The bottom panels show magnified images of the white-squared dorsomedial region in the top panels, where HugPC neurons project their axons close to the cell bodies of the IPCs. While the top panels are z-stacks of 122 image slices (1 μ m interval) covering the entire brain tissue, the bottom panels are projections of 60 slices centering around IPCs. Similar results were obtained from 5 independent samples of the same genotype. (F) The effect of neuronal silencing confined to the HugPC subpopulation. ** p < 0.01, * p < 0.05 (Mann-Whitney U-test with Bonferroni correction).

Hugin triggers Ca^{2+} elevation and Dilp secretion in larval IPCs via PK2-R1

A previous report showed that IPCs exhibit increased firing rates and intracellular calcium (Ca^{2+}) levels upon stimulation (Kr neisz et al., 2010). This motivated us to next test if Hugin activates IPCs. To further visualize the output pattern of HuginPC neurons, we expressed the presynaptic marker Syt-eGFP in HuginPC neurons (Figure 4—figure supplement 1). To test whether Hugin activates IPCs, we utilized the recently developed calcium indicator CRTC::GFP, which can report Ca^{2+} dynamics in the timescale of several minutes (Bonheur et al., 2023). As glucose reportedly triggers Ca^{2+} dynamics in IPCs in a range of minutes (Oh et al., 2019), we wondered if CRTC::GFP can detect this dynamics, as a positive control. To this end, we applied D-glucose to isolated larval brains and found a significant cytosol-to-nuclear translocation of CRTC::GFP in IPCs within ~5 min, confirming that this reporter can indeed detect Ca^{2+} responses in IPCs upon stimulation (Figure 5B). We then examined whether *Hug* neurons act upstream of PK2-R1-expressing IPCs by measuring Ca^{2+} responses of IPCs while thermogenetically activating *Hug* neurons with TrpA1. We found that activation of *Hug* neurons tended to increase Ca^{2+} levels in IPCs (Figure 4B and C), consistent with the HuginPC-to-IPCs axis (Figure 4D). Next, we tested whether Hug peptides can induce Ca^{2+} responses in IPCs using *ex vivo* preparations. The *Hug* gene encodes a precursor that potentially produces at least two distinct peptides, Hug-  and PK-2, both of which can bind PK2-R1 to induce Ca^{2+} responses (Rosenkilde et al., 2003). We thus incubated the CRTC::GFP-expressing larval brain with chemically synthesized Hug-  or PK-2 (Kr neisz et al., 2010; Oh et al., 2019) (Figure 5A). We found that bath application of either Hug-  or PK-2 induced significant Ca^{2+} responses in IPCs (Figure 5B–D). In agreement with the idea that Hug peptides activate IPCs via PK2-R1, knock-out of PK2-R1 blocked the Hug peptide-evoked Ca^{2+} elevations in IPCs (Figure 5F and G). Similar results were obtained when we bath-applied the synthetic Neuromedin U (NMU), the mammalian homolog of Hugin (Figure 5A and 5G), suggesting the role of Hugin/PK2-R1 signaling may be conserved across species. Furthermore, Dilp3 accumulated in IPCs were significantly reduced after applying Hug peptides, suggesting that Hugin causes IPCs to release Dilp3 (Figure 5E). Collectively, these data suggest that *Hug* neurons activate IPCs via the Hug/PK2-R1 signaling to regulate sleep (Figure 5H).

Hugin/PK2-R1 axis has distinct impacts on wake/sleep control in larvae and adults

Given the differences between larval vs adult sleep in temporal pattern and circadian influence, we next wondered if Hug/PK2-R1/Dilps axis functions in adult sleep as well. As in larvae, PK2-R1 knockout adults showed increased sleep amount compared to the control (Figure 6—figure supplement 1B). In contrast, *Hug* knockout or silencing of HuginPC neurons failed to influence adult sleep (Figure 6A and 6B). This suggests that, unlike larvae, Hug is dispensable for adult sleep. Interestingly, we observed that the expression patterns of PK2-R1 and Hug, and the morphology of HuginPC neurons and IPCs, showed a broadly similar distribution in larvae and adults, although we did not directly track individual neurons across development (Figure 6—figure supplement 2). This implies that the differential roles of Hug in larvae vs adults are likely due to physiological differences in HuginPC neurons and/or IPCs. We thus examined how Hug peptides evoke Ca^{2+} responses in adult IPCs and found that neither Hug-  nor PK-2 could induce Ca^{2+} responses in adult IPCs (Figure 6D and 6E), unlike in larvae. Although PK-2 treatment even led to a modest increase in Dilp3 immunoreactivity in adult IPCs (Figure 6F), the physiological significance of this effect remains unclear; at present, we consider it most likely that PK-2 acts on larval and adult IPCs in a different manner. These data indicate that Hug induces Ca^{2+} responses and Dilp secretion in larval IPCs but not in adult IPCs. Surprisingly, we further found that *Dilp3* or *Dilp5* null mutations reduced sleep amounts in adults, consistent with previous reports (Cong et al., 2015) (Figure 6C), while in larvae, these mutations in fact increased the sleep amounts. These data unexpectedly reveal how the same set of neuronal circuits and molecular signaling therein can evolve over development in a complex manner.

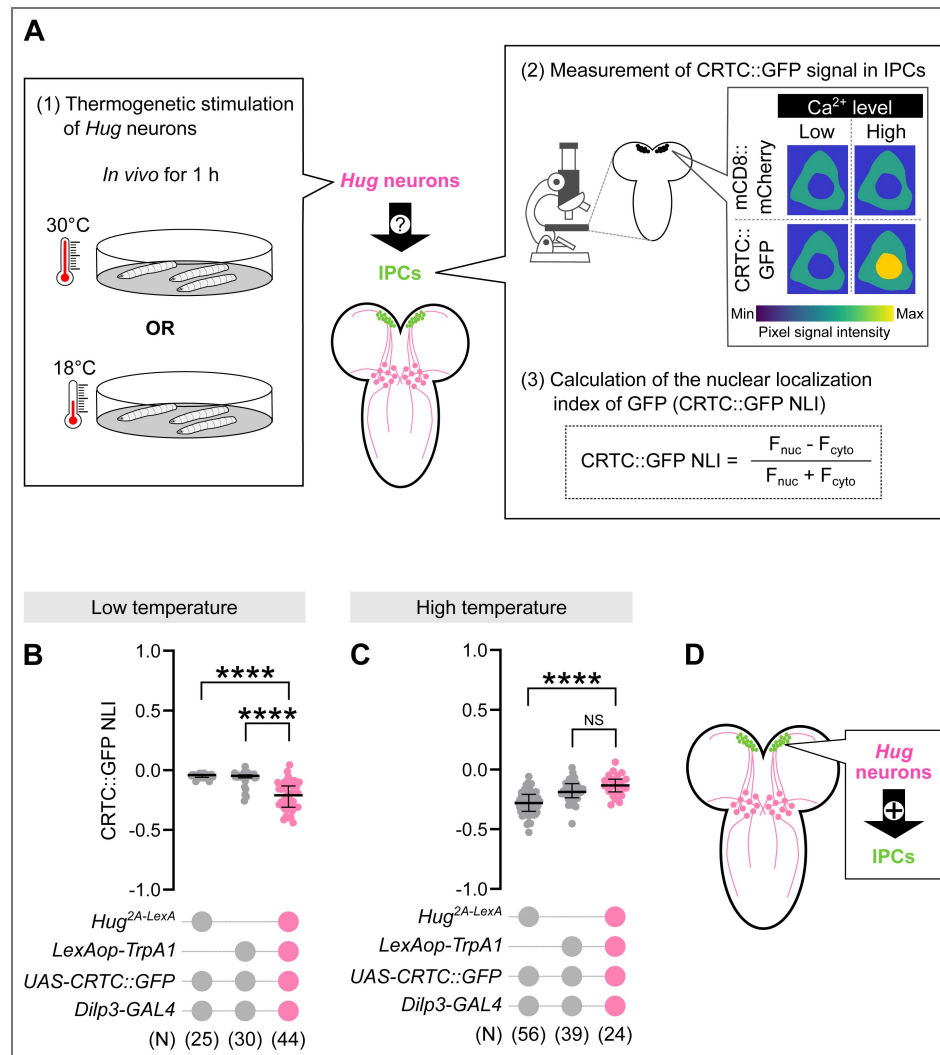


Figure 4. Activation of *Hug* neurons triggers Ca²⁺ responses in larval IPCs.

(A) Schematic workflow for assessing intracellular Ca²⁺ levels in IPCs while thermogenetically activating *Hug* neurons. Larvae were exposed to low (18°C) or high (30°C) temperature for 1 h, followed by imaging of CRTC::GFP in IPCs and calculation of the CRTC::GFP nuclear localization index (NLI; see Methods). (B–C) Nuclear-localized CRTC::GFP signal (CRTC::GFP NLI) in larval IPCs under low-temperature (B) or high-temperature (C) conditions, with or without thermogenetic activation of *Hug* neurons. Each dot represents an individual cell; the thick line indicates the median and the thin error bars indicate the interquartile range (Q1–Q3). “N” indicates the number of cells analyzed per group. **** $p < 0.0001$, NS $p \geq 0.05$ (Mann–Whitney U-test). (D) Working model of a neuronal network in which *Hug* neurons activate IPCs to regulate larval sleep

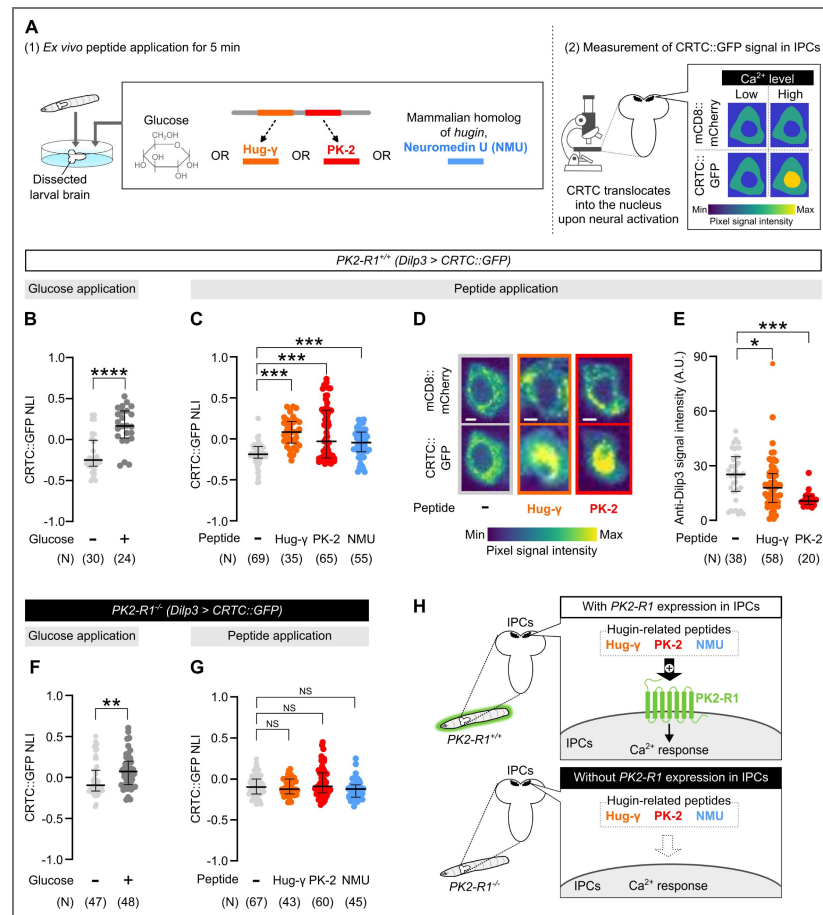


Figure 5. Hug peptides induce Ca²⁺ responses in larval IPCs via PK2-R1

(A) Schematic flow of peptide application followed by Ca²⁺ imaging. (B–C) Ca²⁺ responses in larval IPCs during *ex vivo* bath application of either glucose (B) or Hug peptides (C). In this and the following panels, 'N' indicates the number of cells used for each group, and the thick line and thin error bar represent the median and interquartile range (Q1–Q3), respectively. **** $p < 0.0001$, *** $p < 0.001$ (Mann–Whitney U-test with Bonferroni correction). (D) Representative images of larval IPCs upon peptide application. Scale bars, 2 μ m. (E) Anti-Dilp3 signal intensity measured within the cytosolic areas of larval IPCs. *** $p < 0.001$, * $p < 0.05$ (Mann–Whitney U-test with Bonferroni correction). (F–G) Ca²⁺ responses in larval IPCs derived from PK2-R1 knockout mutants, measured after either glucose (F) or Hug peptide application (G). ** $p < 0.01$, NS $p \geq 0.05$ (Mann–Whitney U-test with Bonferroni correction). (H) A model where Hug peptides, but not glucose, activate IPCs via the PK2-R1 receptor in the larval brain.

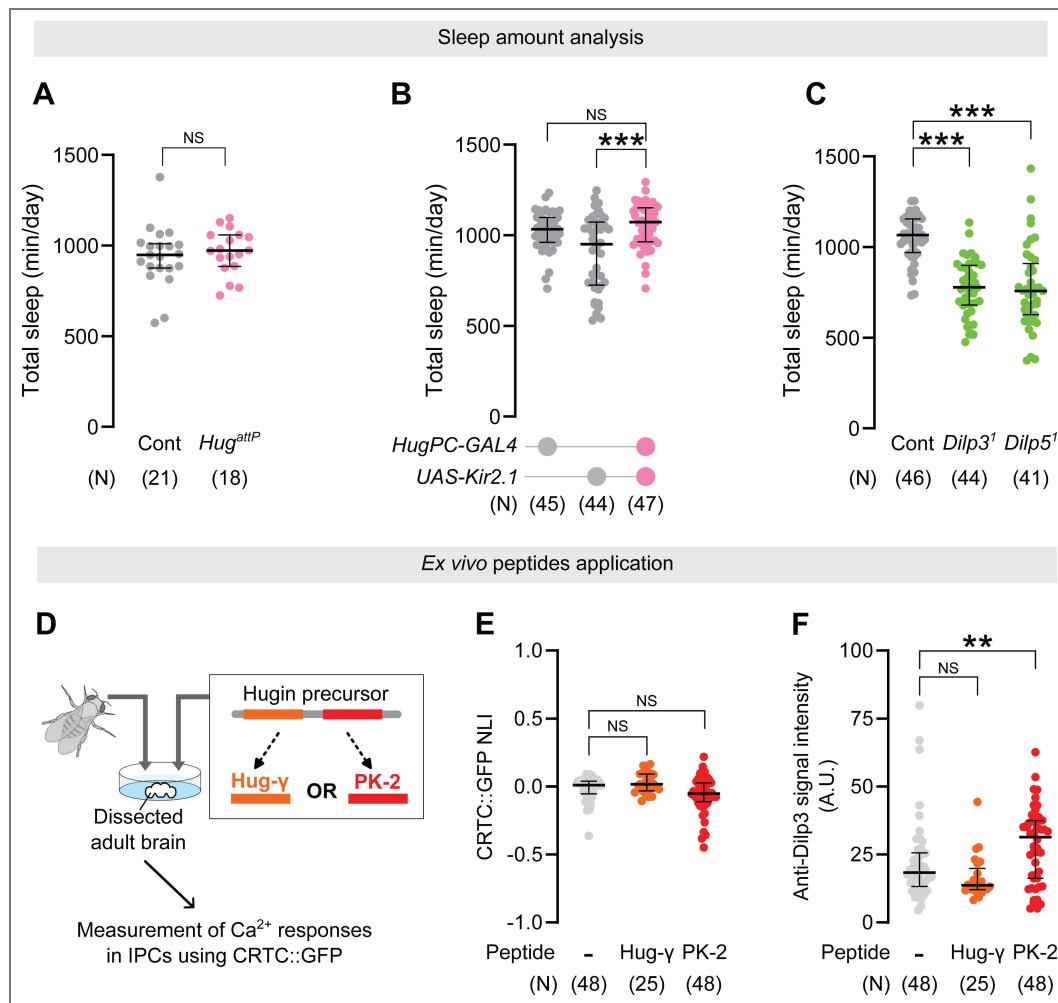


Figure 6. Distinct impacts of Hugin/PK2-R1 axis on wake/sleep control in larvae and adult

(A) Total sleep amounts in *Hug* knockout mutant adults. NS $p \geq 0.05$ (Mann-Whitney U-test). In these and the following panels, 'N' indicates the number of biologically independent animals used for each group, and the thick line and thin error bar represent the median and interquartile range (Q1-Q3), respectively. NS $p \geq 0.05$ (Mann-Whitney U-test). (B) Sleep amounts of adults in which HugPC neurons were silenced. *** $p < 0.001$, NS $p \geq 0.05$ (Mann-Whitney U-test with Bonferroni correction). (C) Sleep amounts of *Dilp3* or *Dilp5* null mutant adults. *** $p < 0.001$ (Mann-Whitney U-test with Bonferroni correction). (D) Schematic flow of peptide application experiments using adult brains. (E-F) Ca²⁺ sponges (E) or anti-Dilp3 signal intensity (F) in adult IPCs after Hug peptide application. ** $p < 0.01$, NS $p \geq 0.05$ (Mann-Whitney U-test with Bonferroni correction).

Discussion

In this study, we have identified the neuropeptide Hugin and its receptor PK2-R1 as a ligand/receptor pair critical for sleep regulation in *Drosophila* larvae. We have further shown that larval IPCs express PK2-R1, respond and release Dilp3 upon Hugin stimulation. Surprisingly, the Hugin/PK2-R1 axis is dispensable for sleep regulation in adults, even though the gene expressions as well as the circuit structure appeared to be conserved between larval and adult brains. Furthermore, Dilps appear to modulate sleep in opposite directions in larvae and adults. Our findings thus uncover the neuropeptidergic modulation circuitry that specifically regulates developmental sleep, and suggest divergent usage of the same molecules and circuitry for sleep modulation in larvae and adults.

Hugin/PK2-R1/Dilps axis negatively regulates sleep in larvae

The present study shows that the neuropeptide Hugin and its receptor PK2-R1 are required for sleep regulation in the early larval stage. This notion is supported by following lines of evidence. First, knock-out mutations in *Hugin* or *PK2-R1* increased sleep amounts compared to the control (Figure 1A and 3A). Consistently, silencing a subpopulation of *Hugin*-expressing neurons, called HugPC neurons, or *PK2-R1* neurons increased sleep amounts (Figure 1B, 3B and 3F). Second, thermogenetic activation of Hugin neurons decreased sleep amounts (Figure 3D). Third, HugPC neurons project their axons to IPCs in the brain that express PK2-R1 receptors (Figure 3E). Fourth, Hugin peptides induced Ca^{2+} elevation and promoted Dilp3 release from IPCs in a PK2-R1-dependent manner (Figure 5). Last, *Dilp3* null mutations increased sleep amounts (Figure 2D). Based on these data, we propose a model in which the Hugin/PK2-R1/Dilps axis along the HugPC-IPC circuitry modulates larval sleep (Figure 7).

Examples of Hugin acting through PK2-R1 are prominent in the context of feeding regulation in larvae and adults (Schoofs et al., 2014). For example, a recent study has suggested that Hugin regulates the timing of pupariation and body wall contraction during pupariation through PK2-R1 in PTTH neurons (Ohhara et al., 2024). In the present study, we showed that HugPC in the larval brain modulates sleep (Figure 3F). A previous electron microscopic analysis suggested that larval HugPC neurons likely form synaptic connections to downstream neurons including IPCs (Hückesfeld et al., 2021; Schlegel et al., 2016). We here showed that PK2-R1 is required for larval sleep control and that thermogenetic activation of Hug neurons is associated with Ca^{2+} responses in PK2-R1-expressing IPCs *in vivo* (Figure 4). Importantly, our *ex vivo* data indicated that Hugin peptide-induced Ca^{2+} response in IPCs was largely abolished by *PK2-R1* mutation (Figure 5), indicating the critical role of the Hugin/PK2-R1 axis for the HugPC-mediated IPC activation. Together, we propose that the HugPC-IPCs circuitry mobilizes Hugin/PK2-R1 signaling to selectively modulate larval sleep. Interestingly, our *ex vivo* data indicated that Neuromedin U (NMU), a mammalian ortholog of Hugin, can trigger Ca^{2+} responses in larval IPCs in a PK2-R1-dependent fashion (Figure 5). Given that NMU has been implicated in sleep regulation in fishes and mammals (Ahnaou & Drinkenburg, 2011; Chiu et al., 2016), an analogous ligand/receptor pair might be involved in sleep modulation in mammals as well.

In this study, activation of Hugin–PK2-R1 signaling by bath application of Hugin peptides led to a significant reduction in Dilp3 immunoreactivity in IPCs, whereas Dilp2 immunoreactivity did not show a decrease under these conditions, suggesting that Hugin–PK2-R1 activation promotes Dilp3 secretion without strongly enhancing Dilp2 release in our assay. One plausible explanation is that individual Dilps are stored in partially distinct vesicular pools within IPCs and are released through stimulus-specific mechanisms. Consistent with this idea, previous work has shown that Dilp release from IPCs can be selective, with Dilp2 and Dilp3 responding to distinct nutritional cues (Kim and Neufeld, 2015). Confocal analyses in that study further demonstrated that Dilp2 and Dilp3 segregate into different intracellular granules, supporting the notion that individual Dilps can be targeted to distinct secretory pathways within the same neurons. In light of these observations, the selective effect of Hugin–PK2-R1 activation on Dilp3, but not Dilp2, is readily interpreted as another example of non-uniform Dilp regulation. Rather than uniformly driving

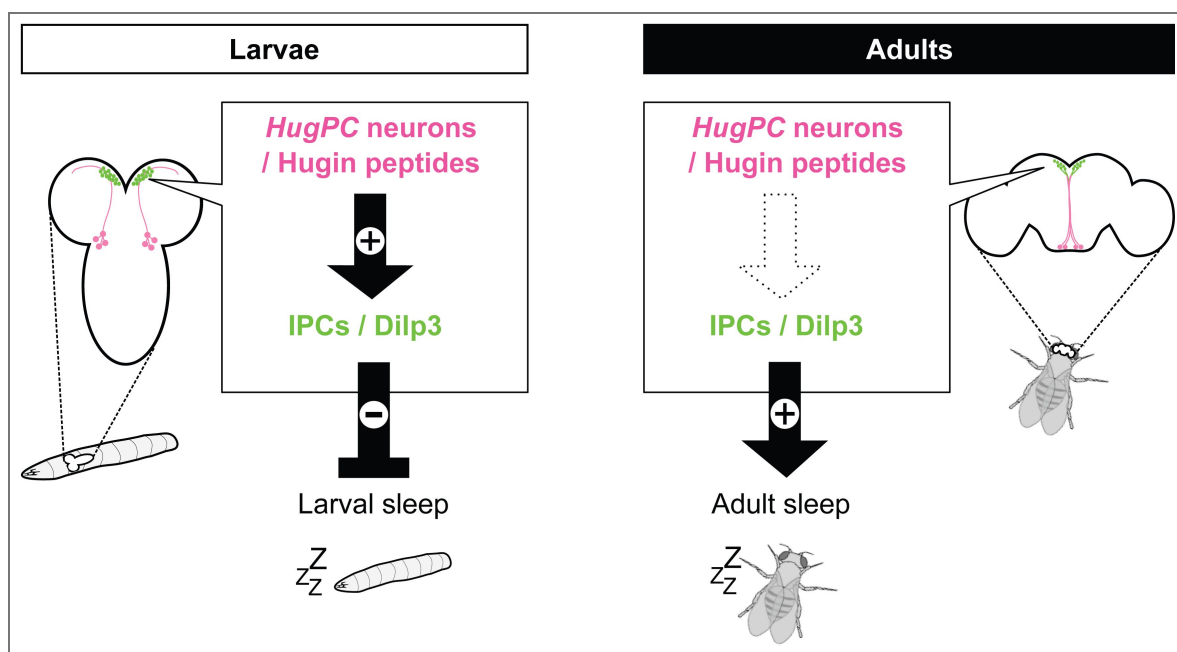


Figure 7. A schematic model of the Hugin/PK2-R1/Dilps axis in larval and adult sleep

Schematic summary comparing larvae and adults. In larvae, bath application of hugin peptides activates IPCs, and both hugin signaling (HugPC neurons/hugin peptides) and IPC output (including Dilp3) act to suppress larval sleep. In adults, hugin peptides do not activate IPCs (dashed arrow), whereas IPC/Dilp3 signaling promotes adult sleep.

secretion of all IPC-derived Dilps, we propose that Hugin–PK2-R1 signaling preferentially mobilizes the Dilp3 secretory pathway. Such selective regulation is reminiscent of piecemeal degranulation in mammalian eosinophils and mast cells, as well as stimulus-dependent mobilization of specific vesicle pools in pancreatic β cells. In this model, different Dilps are stored in partially distinct vesicular pools and are mobilized by different upstream inputs. While ultrastructural approaches such as electron microscopy could provide further insight into the anatomical segregation of Dilp-containing vesicles, these analyses are technically beyond the scope of the present study, and we therefore highlight them as an important direction for future work.

PK2-R1 knockout larvae exhibited both an increase in sleep and a reduction in locomotion speed. Because our locomotor metric is defined as travel distance per unit time during epochs classified as wake, the observed decrease in locomotion speed cannot be fully explained by the increased fraction of time spent in sleep. Nevertheless, we cannot exclude the possibility that part of the apparent sleep increase in the knockout reflects broader motor impairments in PK2-R1–expressing circuits. PK2-R1 expression is not confined to IPCs but is broadly distributed in the nervous system, including populations in the ventral nerve cord (Figure 1C), making it likely that PK2-R1–positive neurons outside IPCs contribute to locomotor control. Consistent with this idea, when we restricted our manipulations to IPCs, we observed robust changes in sleep without significant alterations in locomotor activity (Figure 2—figure supplement 1). These findings suggest that IPCs are more directly involved in sleep regulation, whereas other PK2-R1–expressing neurons may predominantly influence locomotion.

Dh44 neurons promote arousal in second-instar larvae (Poe et al., 2023), raising the possibility that hugin signaling may influence sleep through Dh44 in addition to IPCs. Consistent with this idea, the hugin receptor PK2-R1 is expressed not only in IPCs but also in Dh44 and DMS neurons (Schlegel et al., 2016). Thus, an extension of our model is that HuginPC-derived hugin may act on multiple PK2-R1–positive targets (including IPCs and potentially Dh44/DMS neurons) to shape larval sleep. Future cell type-specific perturbations of PK2-R1 in Dh44 and DMS neurons will be important to test the contribution of these pathways.

Divergent roles of insulin signaling in regulation of larval sleep and feeding

IPCs in larvae respond to exogenously applied Hug peptides, both by exhibiting elevated Ca^{2+} levels and decreases in intracellular Dilp3 storage (Figure 5C–E). In contrast to Dilp3, however, the amount of Dilp2 accumulated in IPCs remains unchanged following Hug peptide application (Figure 5—figure supplement 1B). On the other hand, Dilp2 accumulation in IPCs decreases after glucose treatment (Figure 5—figure supplement 1A) (Oh et al., 2019). These results unexpectedly suggest that larval IPCs form parallel channels inside, one for Hug–Dilp3 and the other for glucose–Dilp2. Mechanistically, Hug acts directly on IPCs via the PK2-R1 receptor. As for glucose, on the other hand, glucose-sensing CN neurons in the brain take up glucose and subsequently secrete another neuropeptide to activate IPCs (Oh et al., 2019). Such parallelism inside IPCs might provide insight into why insect species broadly express multiple insulin-like peptide genes, in some cases even as many as 40 genes in the silkworm *Bombyx mori* (Okamoto, 2021). *Drosophila* genome harbors eight Dilp genes (Nässel et al., 2015), of which Dilp2, Dilp3, and Dilp5 are considered nutrient-related. Nevertheless, mRNA expression or release patterns of these Dilps are not identical under different nutrient conditions (Géminard et al., 2009; Kauffman & DiAngelo, 2024; Nässel et al., 2015; Oh et al., 2019; Tomoatsu et al., 2002). Furthermore, these Dilps exhibit different binding affinities for the insulin receptor (InR), the secreted decoy of InR (SDR), the ecdysone-inducible gene L2 (Imp-L2) (Okamoto et al., 2013). These reports hint at the differential roles of different insulin-like peptides, but mechanistic insights remain largely lacking. Our data of larval IPCs forming parallel channels for Hug–Dilp3 and glucose–Dilp2 may provide a point of entry for tackling this mystery.

Differential impacts of the same neuropeptidergic modulations on larval and adult sleep

Our data show that the Hugin/PK2-R1/Dilps axis likely downregulates sleep amounts in larvae. Despite the significant increase of sleep amount in *Hugin* mutant larvae, *Hugin* mutant adults showed sleep amounts comparable to wild-type control (Figure 6A [↗](#)), consistent with a previous report that Hugin is dispensable for control of baseline sleep levels (Schwarz et al., 2021 [↗](#)). According to our histological analyses, the Hugin/PK2-R1 expression patterns as well as the HugPC-IPCs circuitry appear to be largely conserved in adults (Figure 3E [↗](#) and Figure 6—figure supplement 2 [↗](#)). Unlike larval brain, however, Hugin peptides failed to trigger Ca^{2+} responses in IPCs in adults (Figure 6E [↗](#)). It is thus possible that Hugin/PK2-R1 signaling along the HugPC-IPCs circuitry is suppressed in adults. IPCs in adults receive multiple positive and negative modulatory inputs through GPCRs including the metabotropic GABA_B receptors (Enell et al., 2010 [↗](#)), which suppresses IPC activity and Dilp release in adult IPCs (Enell et al., 2010 [↗](#)). It is thus plausible that such negative modulatory inputs to IPCs in adults might counteract with the Hugin/PK2-R1 axis to suppress Dilp release. In addition, our data suggest that Dilps modulate sleep amount in the opposite directions in larvae and adults (Figure 7 [↗](#)). Comparing the expression levels and activities of GPCRs in larval and adult IPCs would be essential to better understand how same modulatory signals over the course of development come to exert differential impacts on sleep. Interestingly, Hugin in adults appears irrelevant for the baseline sleep amount but is required for homeostatic regulation of sleep (Schwarz et al., 2021 [↗](#)). Thus, testing if Hugin/PK2-R1 axis is involved in the homeostatic regulation of larval sleep, and how such a system compares to its adult counterpart, may further provide mechanistic insights into how homeostatic sleep regulation matures over development.

In summary, we identified the Hugin/PK2-R1/Dilps signaling along the HugPC/IPCs circuit as the molecular and circuitry basis for developmental sleep regulation. Our study begins to fill the knowledge gap in larval sleep regulation, and sheds light on mechanistic differences in sleep regulation between early developmental stages and adults.

Materials and Methods

Fly strains

The following strains of *Drosophila melanogaster* were obtained from the Bloomington *Drosophila* Stock Center: *iso31* (an isogenic w^{1118} strain; BDSC# 5905), *PK2-R1^{attP}* (BDSC# 84563), *PK2-R1^{2A-GAL4}* (BDSC# 84686), *PK2-R1^{2A-LexA}* (BDSC# 84431), *Dilp3¹* (BDSC# 30882), *Dilp5¹* (BDSC# 30884), *Dilp3-GAL4* (BDSC# 52660), *Dilp5-GAL4* (BDSC# 66007), *Hug^{attP}* (BDSC# 84514), *Hug^{2A-GAL4}* (BDSC# 84646), *Hug^{2A-LexA}* (BDSC# 84397), *UAS-Kir2.1::EGFP* (BDSC# 6596), *UAS-Stinger*, *LexAop-tdTomato.nls* (BDSC# 66680), *UAS-mCD8::mCherry*, *UAS-CRTC::GFP* (BDSC# 99657), *UAS-ReaChR* (BDSC# 53741), *UAS-TrpA1* (BDSC# 26263), *LexAop-ReaChR* (BDSC# 53746), *LexAop-rCD2::RFP-p10.UAS-mCD8::GFP-p10* (BDSC# 67093). w^{1118} was from Y. Rao (Peking University); *HugPC-GAL4* (Hückesfeld et al., 2016 [↗](#)) was from M. Pankratz (University of Bonn); *LexAop-TrpA1* (Burke et al., 2012 [↗](#)) was from S. Waddel (University of Oxford). *HugPC-LexA* was created in this study (see the following section, “Generation of transgenic lines”). Detailed lists of fly genotypes can be found in the key resource table and Source Data files.

General fly maintenance

Flies were maintained in an incubator (PHCbi, model# MTR-554-PJ) set to 25°C with a humidity of over 60% and a 9AM:9PM light/dark cycle. After 10 d of rearing in bottles, old flies were replaced with freshly eclosed ones to maintain the strain (Furusawa et al., 2023 [↗](#)).

Egg collection

Agar substrates for egg collection were prepared by microwave-heating a mixture of 2.0 g sucrose (Fujifilm Wako Pure Chemicals, cat# 196-00015), 3.0 g agar (BD, cat# 214010), 2.5 mL apple juice (Dole), and Milli-Q water up to 100 mL. Melted agar was poured into Petri dishes with a diameter of 4 cm (approximately 3 mL per dish). Solidified agar was scratched with tweezers to make the surface rough. Agar plates were stored inside a sealed container at 4°C. Upon egg collection, yeast paste (dried yeast (Nippon Beet Sugar Co., Ltd.) kneaded with Milli-Q water at 40% (w/w)) was placed in the center of the agar plates to further promote oviposition.

Larval staging

First-instar larvae were collected 2 d after egg collection. Among them, second-instar larvae that molted within a 2-h window were used for video recording. Distinguishing between first- and second-instar larvae was based on morphological characteristics: while anterior spiracles were not evident in the first-instar, fist-shaped spiracles were observed in the second-instar (Lakhotia and Ranganath, 2021 [↗](#)).

Larval sleep analysis

Twenty-four-well silicone chambers for sleep analysis, described in Figure 1—figure supplement 1 [↗](#), were made by pouring dimethylpolysiloxane (PDMS) into molds based on a previous study (Churgin et al., 2019 [↗](#); Szuperak et al., 2018 [↗](#)). As a substrate, 120 μ L of a mixture of 3% agar and 2% sucrose was poured into each well. After the gel solidified, 10 μ L per well of 60 mg/mL yeast suspension was applied to the surface of the gel. Larvae were placed individually into the wells, and a glass plate (Azone, cat# 1-4540-01, 5 mm $\times$ 200 mm $\times$ 200 mm) was positioned on top of the chamber to prevent larvae from escaping. The chamber was placed inside a dark box equipped with a ring-type infrared LED (CCS, cat# LDR2-132IR940-LA) and a camera (The Imaging Source, cat# DMK27BUP031), and recording was started immediately without acclimation. Images were captured through the IC Capture software (The Imaging Source, version 2.5) under the following conditions: MJPEG Compressor for codec, a frame rate of 0.87 fps, and an image size of Y800 (2592 $\times$ 1944). Locomotor activity was quantified from the same video recordings used for larval sleep analysis. Travel distance was calculated as the cumulative displacement of the larval centroid over time, and locomotion speed was defined as total travel distance divided by the total duration of wakefulness, considering only time bins classified as wake. To minimize the possibility that feeding behavior was misclassified as sleep, we adjusted our immobility threshold using video recordings in which larval feeding bouts, including relatively subtle episodes, were apparent, and configured the algorithm so that feeding-associated movements would generally be classified as wake rather than inactivity. During the development and routine use of the assay, we also inspected representative recordings across genotypes and did not observe obvious feeding-related abnormalities or gross locomotor defects under our HuginPC/IPC manipulations.

Sleep deprivation

Blue light irradiation was performed using a 455-nm LED (Thorlabs, cat# M455L4) with a collimating adapter (Thorlabs, cat# SM1U25-A) (Omamiuda-Ishikawa et al., 2020 [↗](#); Yoshino et al., 2017 [↗](#); Yoshino et al., 2025 [↗](#)). The irradiation intensity was set to the maximum scale using a T-cube LED driver (Thorlabs, cat# LEDD1B). Temporal control of LED illumination was achieved using the Arduino UNO microcontroller board (Arduino, *RRID:SCR_017284* [↗](#)). For sleep deprivation experiments, the first 1 h was recorded without LED illumination, followed by another 1 h consisting of repeated cycles of “LED ON for 90 s and OFF for 30 s”.

Immunohistochemistry

Antibody staining of larval neurons was carried out as previously reported (Morikawa et al., 2011 [↗](#); Omamiuda-Ishikawa et al., 2020 [↗](#)). Briefly, dissection in phosphate-buffered saline (PBS); fixation in 4% paraformaldehyde/PBS at room temperature for either 90 min in CRTC::GFP

experiments or 30 min in the others; washing with 0.3% Triton X-100/PBS (hereafter referred to as PBST) for 10 min $\times$ 3; blocking in 5% normal goat serum (NGS)/PBST at room temperature for 30 min; primary antibody treatment at 4°C overnight; washing with PBST for 10 min $\times$ 3; secondary antibody treatment at 4°C overnight; washing with PBST for 10 min $\times$ 3; and mounting in VECTASHIELD medium (Vector Laboratories, cat# H-1000, *RRID:AB_2336789* [↗](#)). Reactions were performed either in Terasaki 96-well plates (stem Corp., cat# P96R37N) for adults or in 48-well culture plates (Corning, cat# 3548) for larvae. The following primary antibodies were diluted in 5% NGS/PBST: chicken polyclonal anti-GFP (1:1000, Aves Labs, cat# GFP-1020, *RRID:AB_10000240* [↗](#)), mouse monoclonal anti-mCherry (1:1000, Takara Bio, cat# 632543, *RRID:AB_2307319* [↗](#)), rabbit polyclonal anti-Dilp3 (1:250, (Veenstra et al., 2008 [↗](#))), rabbit polyclonal anti-Dilp2 (1:2000, (Okamoto et al., 2012 [↗](#))), mouse monoclonal anti-BRP (1:100, Developmental Studies Hybridoma Bank, clone nc82; *RRID:AB_2314866* [↗](#)), and rat monoclonal anti-histone H3 phospho S28 (1:1000, Abcam, cat# ab10543; *RRID:AB_2295065* [↗](#)). The following secondary antibodies were diluted in 5% NGS/PBST: goat anti-chicken Alexa Fluor 488 (1:200, Thermo Fisher Scientific, cat# A-11039, *RRID:AB_2534096* [↗](#)), goat anti-mouse Alexa Fluor 555 (1:200, Thermo Fisher Scientific, cat# A-21424, *RRID:AB_141780* [↗](#)), goat anti-rabbit Alexa Fluor 633 (1:200, Thermo Fisher Scientific, cat# A-21071, *RRID:AB_2535732* [↗](#)), and goat anti-rat Alexa Fluor 633 (1:200, Thermo Fisher Scientific, cat# A-21094; *RRID:AB_2535749* [↗](#)). Fluorescence images were obtained using a confocal microscope (Leica, TCS SP8) with 20 $\times$ (Leica, HC PL APO CS2 20 $\times$ /0.75 IMM) and 63 $\times$ (Leica, HC PL APO CS2 63 $\times$ /1.40 OIL) objectives. Images were captured at 512 $\times$ 512-pixel resolution with an interval of 1 μ m. Z-stack images of the maximum projections were generated using the ImageJ v1.53t software (Schneider et al., 2012 [↗](#); *RRID:SCR_003070* [↗](#)).

Generation of transgenic lines

HugPC-LexA lines were generated by Φ C31 integrase-mediated transgenesis, as previously described (Groth et al., n.d.).

The 544-bp enhancer region of *Hug*, essentially the same as that used to generate *HugPC-GAL4* (Hückesfeld et al., 2016 [↗](#)), was PCR-amplified from fly genomic DNA. The gel-purified amplicon was subjected to another round of PCR to add hangout sequences. The backbone suicide-vector pBPnlsLexA::p65Uw (*RRID:Addgene_26230* [↗](#)) was initially amplified by transforming the *ccdB*-resistant *E. coli* strain, One Shot™ *ccdB* Survival™ 2 T1^R competent cells (Thermo Fisher Scientific, cat# A10460). The PCR product and the vector were double-digested by Aat II (NEB, cat# R0117S) and Fse I (NEB, cat# R0588S), followed by ligation using In-Fusion HD Cloning Kit w/Cloning Enhancer (Takara Bio, cat# 639635). The integrity of the resulting plasmid was confirmed by DNA sequencing. Plasmid insertion was targeted to four attP sites (attP2, attP40, VK00005, and su(Hw)attP5) individually through Φ C31 integrase-mediated transgenesis by BestGene Inc. (Chino Hills, CA). Transformants were selected based on the eye color marker. Among the generated strains harboring *HugPC-LexA* in each insertion site, labeling patterns in the larval brain were checked by driving fluorescent marker expression under the control of *LexAop* (Figure 3—figure supplement 2 [↗](#)). As the *HugPC-LexA* inserted in attP2 closely resembled the labeling pattern of the original *HugPC-GAL4*, it was used in further experiments.

The following primers were used for cloning:

Forward primer for the *HugPC* enhancer; 1st round PCR against genomic DNA:

5'-AAGGGTTTGGTTTAATTTATTTATGTCATA-3'

Reverse primer for the *HugPC* enhancer; 1st round PCR against the genomic DNA:

5'-GGACAACTGATGCCAGCAGC-3'

Forward primer for the *HugPC* enhancer with hangout sequences (underlined); 2nd round PCR against the 1st round amplicon:

5'-gaaaagtgccacctgacgtAAGGGTTTGGTTTAATTTATTTATGTCATA-3'

Reverse primer for the *HugPC* enhancer with hangout sequences (underlined); 2nd round PCR against the 1st round amplicon:

5'-cccgggcgagctcgccggGGACAACTGATGCCAGCAGC-3'

Food intake assay

Food intake was measured using a dye-based feeding assay adapted from Hückesfeld et al. (2016). Apple juice agar plates were prepared with a spot of red yeast paste in the center. After 30 min starvation, larvae were transferred onto the yeast paste and videotaped for 20 min. Larvae were then collected in a small cell strainer and rinsed with 60°C hot water to remove external dye, transferred to glass slides, and photographed. Images were analyzed in ImageJ by measuring the area of red-stained dye signal and the total body area, and food intake was quantified as the ratio of dye-stained area to total body area for each larva.

Measurement of neuronal activity and Dilp3 accumulation in IPCs

The GFP-tagged CRTC (CRTC::GFP) reporter was applied to quantify the activities of larval IPCs, essentially according to the previous report in adult flies (Bonheur et al., 2023).

Parental *Dilp3-GAL4* and the reporter line harboring *UAS-CRTC::GFP* and *UAS-mCD8::mCherry* (BDSC# 99657) were crossed, and eggs were collected within an 8–10 h time window. Offspring larvae were reared for 4 d after egg laying (AEL) at 25°C to early third instar. Collected larvae were washed with water and starved on water-soaked paper for 12–14 h. For *in vivo* thermogenetic activation, larva-containing Petri dishes were warmed at 30°C for 60 min. For *ex vivo* thermogenetic experiments, larvae were dissected in calcium- and sugar-free HL3.1 buffer (Nakamizo-Dojo et al., 2023) at 30°C, and each brain was incubated in the same buffer at 30°C for 5 min. For *ex vivo* glucose or peptide application assays, larvae were first dissected in the same buffer as above at room temperature, and brain samples were then treated with either 20 mM D-glucose or 1 mM of chemically synthesized peptides dissolved in HL3.1 buffer for 5 min. After incubation, samples were subjected to immunohistochemical analyses to visualize GFP, mCherry, and Dilp3 signals (see the section “Immunohistochemistry”). Image processing was performed using ImageJ v1.53t (Schneider et al., 2012; RRID:SCR_003070). Nuclear ROIs and cell body outlines were drawn based on the mCD8::mCherry fluorescence insensitive to neuronal activity, and the mean intensities of CRTC::GFP signals within the nucleus (F_{nuc}) and the cytoplasm ($F_{\text{cyto}} = F_{\text{whole cell}} - F_{\text{nuc}}$) were quantified. NLI was calculated using the following formula: $(F_{\text{nuc}} - F_{\text{cyto}})/(F_{\text{nuc}} + F_{\text{cyto}})$. To assess Dilp3 accumulation in IPCs, the anti-Dilp3 signal intensity was measured within the cytosol of each cell, excluding the nucleus. Data from 4–14 Dilp3-positive IPCs per brain, collected from 3–7 larvae, were pooled for each genotype/treatment group.

We attempted to record GCaMP-based calcium dynamics in Hugin neurons and IPCs, but did not pursue this approach further because it was low-throughput and a positive-control assay (glucose application) did not elicit the expected IPC calcium response in our hands.

Chemical synthesis of Hug peptides

Peptides with the following amino acid sequences were chemically synthesized using the Fmoc solid phase method, and their purity was assessed by HPLC to be over 95%. All procedures were carried out by Eurofins Genomics K.K. (Tokyo, Japan).

Drosophila Hug-γ:

Acetyl-QLQSNGEPAYRVTPRL-CONH₂ (17 a.a.)

Drosophila PK-2:

Acetyl-SVPFKPRL-CONH₂ (8 a.a.)

Human NMU:

Acetyl-FRVDEEFQSPFASQSRGYFLFRPN-CONH₂ (25 a.a.)

Sleep quantification in adult flies

The following parts of the *Drosophila* activity monitor (DAM) system were purchased from TriKinetics Inc. (Waltham, MA): the *Drosophila* Activity Monitor (cat# DAM2), the power supply interface unit (cat# PSIU9), monitor tubes with an outside diameter of 5 mm and a length of 65 mm (cat# PGT5x65), and tube caps (cat# CAP5-Black). Newly eclosed adult flies were collected under brief CO₂ anesthesia and kept as groups of 10–15 in standard food vials. Males and females were kept separately to avoid mating. To minimize disturbance by noise and vibration, flies were placed inside a fan-less Peltier-type incubator (Mitsubishi Electric Engineering Co., Ltd., cat# SLC-25) set to 25°C, externally connected with 5-mm diameter LED lights (MY-CRAFT, Ltd., cat# 5W0601) and a timer device (Panasonic Corp., cat# WH3311WP) to generate a 9AM:9PM light/dark cycle. After 3 d of eclosion, each fly was gently aspirated into a Pyrex glass monitor tube, one end of which was stuffed with agar food (2% (w/w) agar, 5% (w/w) sucrose), and the other was plugged with a cotton string. The tubes were set to the monitor device and placed inside the same fan-less incubator. The DAM system was connected to an external PC installed with DAMSystem3 Data Collection Software (RRID:SCR_021809 [↗](https://doi.org/10.1093/RRID/SCR_021809); available at <https://trikinetics.com/> [↗](https://doi.org/10.1093/RRID/SCR_021809)) for data acquisition, and beam-crossing by each fly was counted for two successive days. Time windows longer than 5 min without beam-crossing were judged as sleep bouts, following the conventional criteria for adult fly sleep (Paul J. Shaw, Chiara Cirelli, Ralph J. Greenspan, & Giulio Tononi, 2000 [↗](https://doi.org/10.1016/S0022-2950(00)00100-0)).

Statistical analysis

Statistical analyses by Kruskal–Wallis one-way ANOVA, Mann–Whitney U-test, or chi-square test were performed using Prism 9.5.1 (GraphPad, RRID:SCR_002798 [↗](https://doi.org/10.1093/RRID/SCR_002798)). Bonferroni correction was applied for multiple comparisons. Asterisks (*) represent p-values as indicated within each figure legend. All data necessary to reproduce the figure panels and statistical analyses are available as Source Data files.

Figure supplements

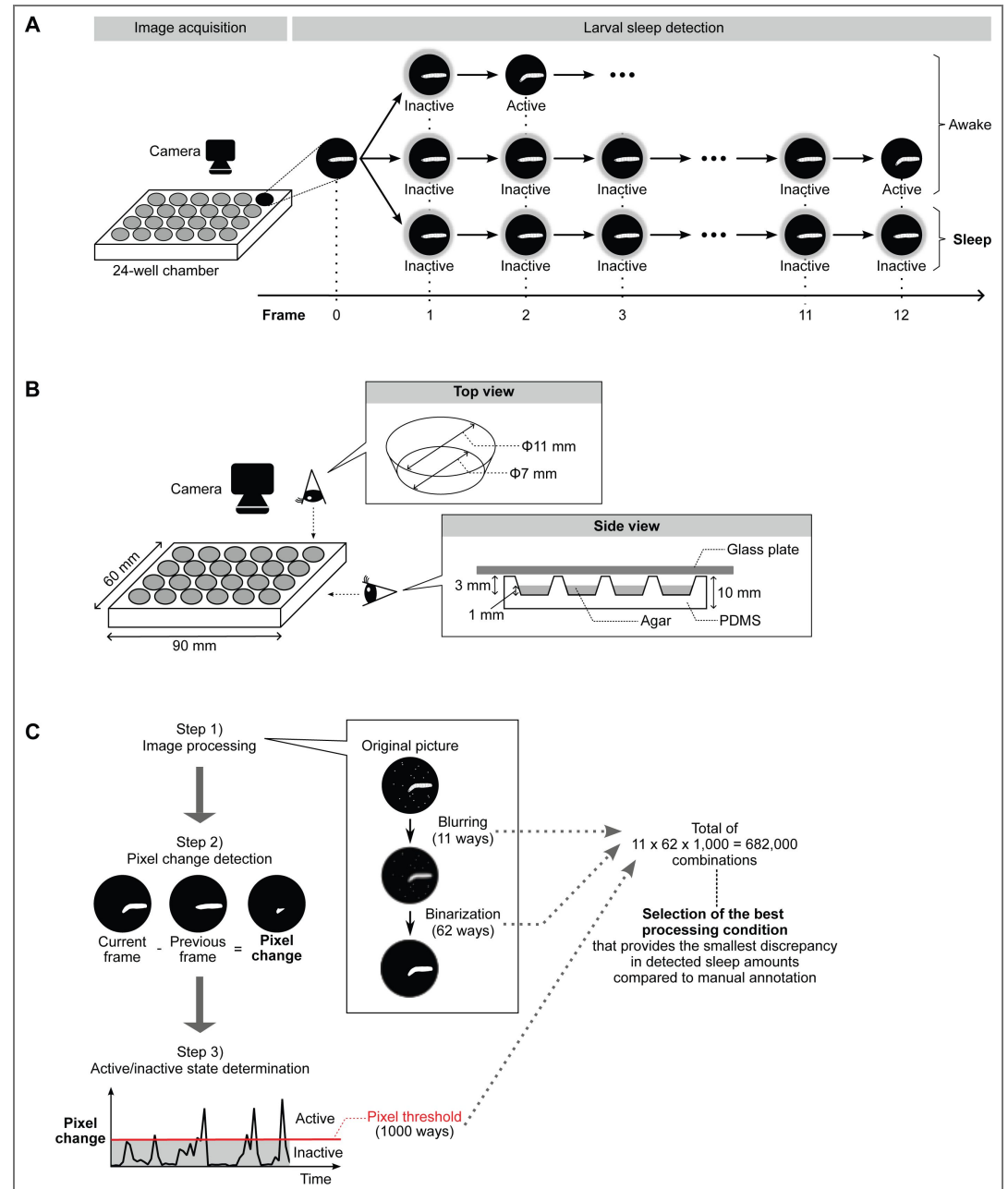


Figure 1—figure supplement 1. Detailed experimental setup for analyzing larval sleep (A) Schematic flow of automated larval sleep quantification. A second instar larva was placed in each of 24 wells, and all 24 wells were videotaped from above. For each well and frame, the image of larva was blurred and binarized to automatically locate the larva. Consecutive frames were then compared so that changes in pixel values were summed to estimate larval motion. This motion estimate was then used to classify every consecutive pair of frames as representing “active” or “inactive.” We found that inactivity across 12 consecutive frames meets the sleep criteria, as shown in Figure 1—figure supplement 2. (B) Schematic representation of the 24-well chamber used for larval sleep assays. (C) Flow of automated processes of larval motion detection and active/inactive state determination. For each well and frame, the image of larva was blurred and binarized. Consecutive frames were then compared so that changes in pixel values were summed to estimate larval motion. This motion estimate was then used to classify every consecutive pair of frames as representing “active” or “inactive.”

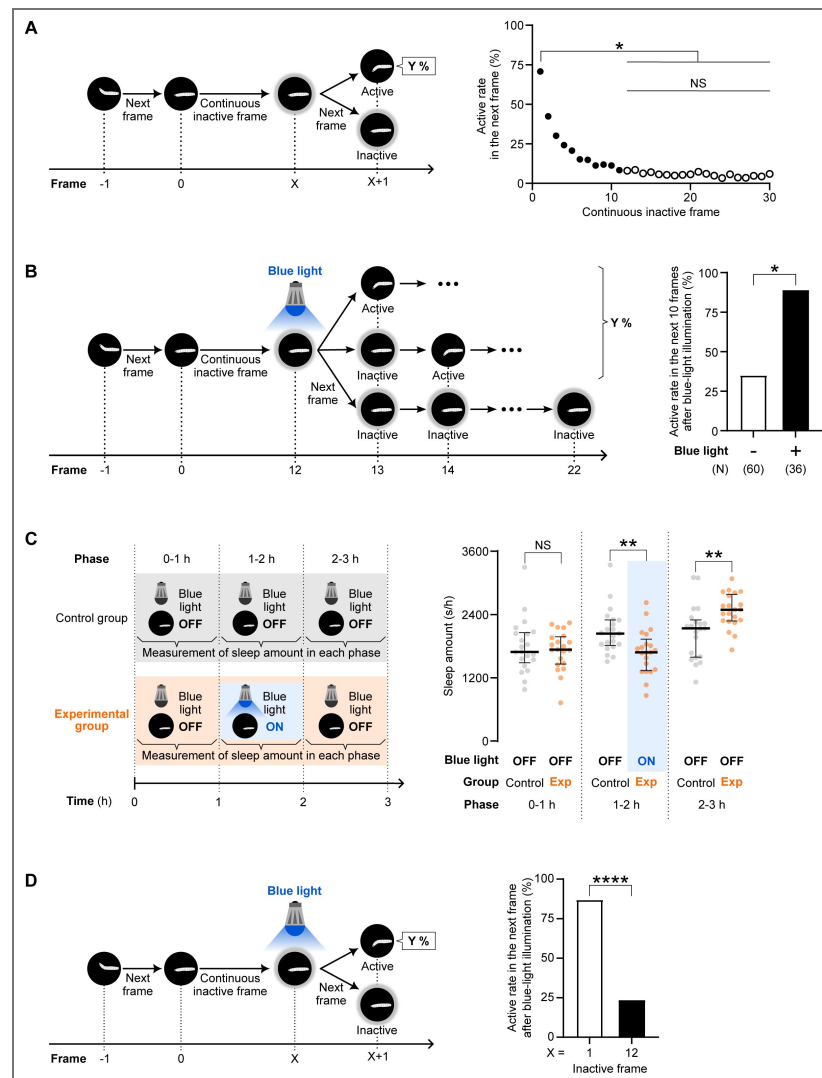


Figure 1—figure supplement 2. Larval “sleep state” in this study is consistent with behavioral criteria for sleep

(A) Relationship between “inactive” state duration and spontaneous arousal probability. Sustained quiescence for ≥ 12 frames (open circles) significantly reduced arousal probability compared to 1-frame inactivity. * $p < 0.05$, NS: $p \geq 0.05$ (chi-squared test). (B) The percentage of larvae that showed a response within the next 10 frames after high-intensity blue light. The number of events analyzed was 36 and 60 with and without blue light irradiation, respectively. * $p < 0.05$ (Fisher’s exact test). (C) Sleep amounts before, during, and after blue light irradiation. The number of larvae tested was 21 and 21 for the control and experimental groups, respectively. ** $p < 0.01$, NS: $p \geq 0.05$ (Mann-Whitney U-test with Bonferroni correction). (D) Percentage of larvae that showed a response in the next 1 or 12 frames after blue light stimulation. Number of events analyzed: 1960 or 111 for 1- or 12-frame continuous inactivity prior to blue light illumination, respectively. **** $p < 0.0001$ (chi-squared test).

Figure 1—figure supplement 3. CRISPR-knock-out screen for genes that regulate larval sleep.

Total sleep amounts in CRISPR-knock-out mutants were measured for 18 h from the beginning of the second instar. The data for Control, *PK2-R1*, and *Oamb* mutants are identical to those shown in Figure 1. Box plots are generated so that the center line indicates the median, the box limits indicate the upper and lower quartiles, and the whiskers indicate the minimum-to-maximum range. *** $p < 0.005$ (Kruskal–Wallis one-way ANOVA and post hoc Mann–Whitney U-test with Bonferroni correction). Number of larvae tested: 475 for the control and 10–52 for each mutant.

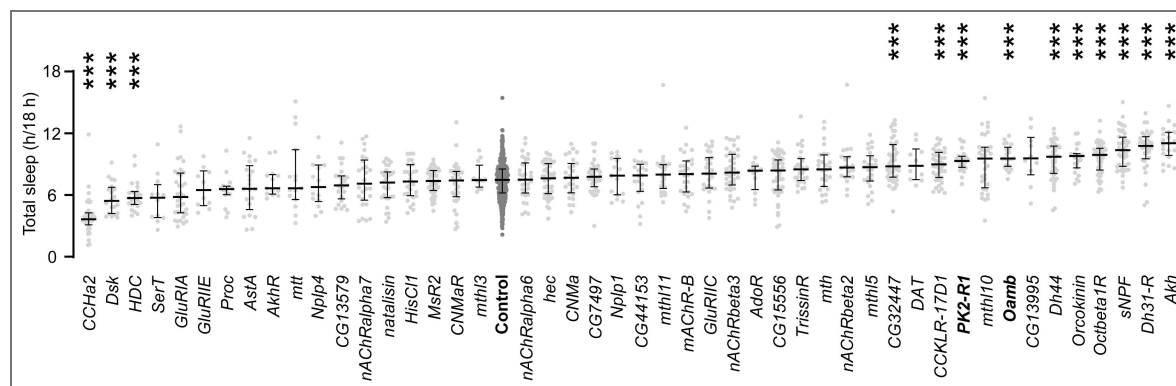
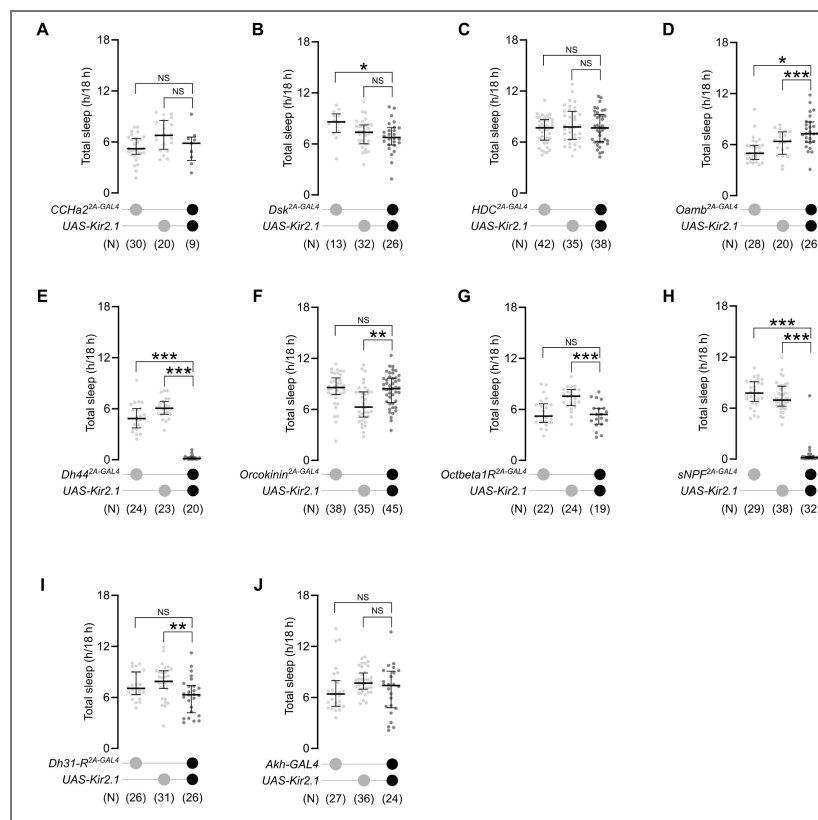


Figure 1—figure supplement 4. Effects of neuronal manipulation on larval sleep

Sleep amounts in larvae in which distinct neuronal populations were silenced. Kir2.1 expression was driven by each enhancer-GAL4 for “hit” genes identified in the CRISPR-knock-out screen. Each dot represents an individual animal in these panels, “N” indicates the number of biologically independent animals per group, and the thick line and thin error bars indicate the median and interquartile range (Q1–Q3), respectively. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, NS: ≥ 0.05 (Mann–Whitney U-test with Bonferroni correction)



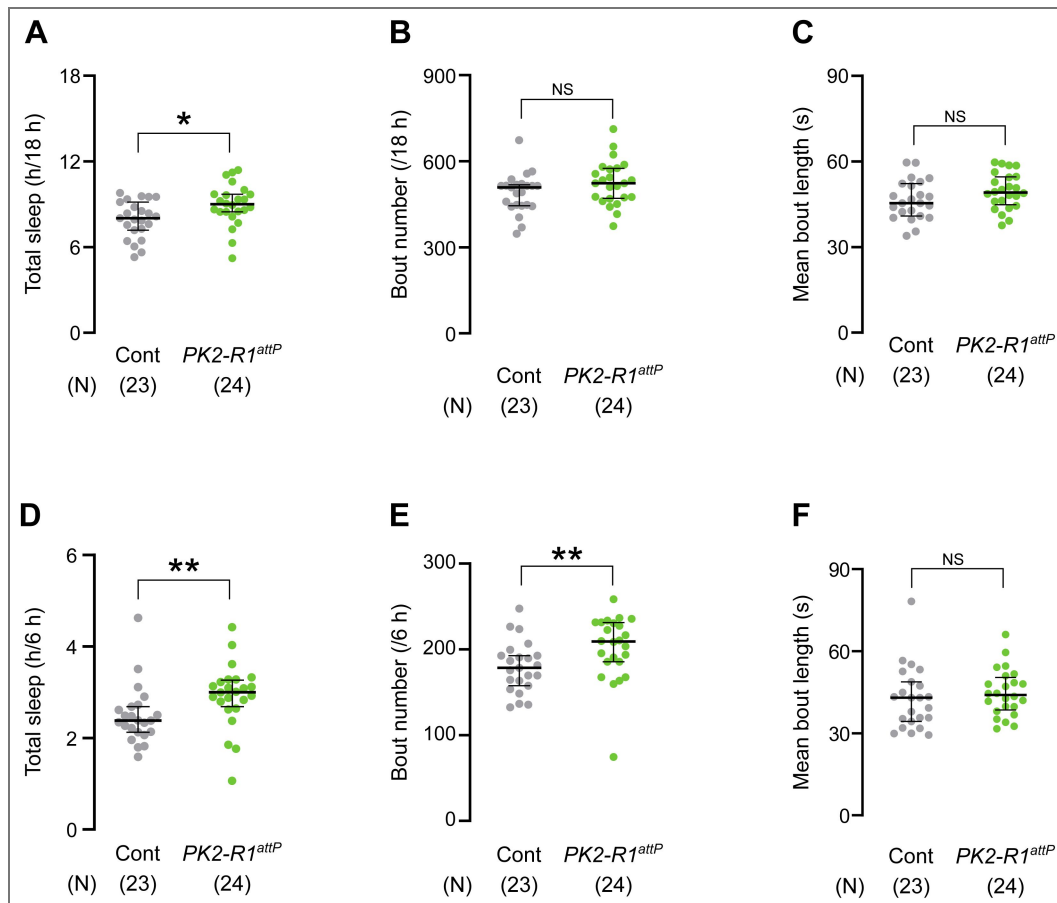


Figure 1—figure supplement 5. Larval sleep phenotypes in PK2-R1 mutants quantified over 18 h and the first 6 h.

(A–C) Sleep metrics quantified over an 18-hour period in second-instar larvae, whereas (D–F) show the same metrics quantified over the first 6 hours in the same animals. Total sleep is shown in (A) and (D), bout number in (B) and (E), and mean bout length in (C) and (F). Each dot represents an individual larva; the thick line indicates the median and the thin error bars indicate the interquartile range (Q1–Q3). Numbers in parentheses indicate N, the number of biologically independent animals per group. ** $p < 0.01$, * $p < 0.05$, NS: ≥ 0.05 (Mann–Whitney U-test with Bonferroni correction).

Figure 2—figure supplement 1. Locomotion speeds are not consistently affected by genetical manipulations of *PK2-R1*, IPCs, or *Dilps*

(A) Larval locomotion speed in controls and *PK2-R1* mutants. (B-C) Effects of silencing *PK2-R1* neurons (B) and IPCs (C) on larval locomotion speed, shown together with the corresponding genetic controls. (D) Larval locomotion speed in *Dilp3* or *Dilp5* null mutants. Average locomotion speed during wake periods was calculated from the datasets shown in Figures 1A, 1B, 2C, and 2D. N indicates the number of biologically independent larvae per group. Each dot represents an individual larva; thick lines indicate the median and thin error bars indicate the interquartile range (Q1-Q3). *** $p < 0.0001$, * $p < 0.05$, NS: $p \geq 0.05$ (Mann-Whitney U-test with Bonferroni correction).

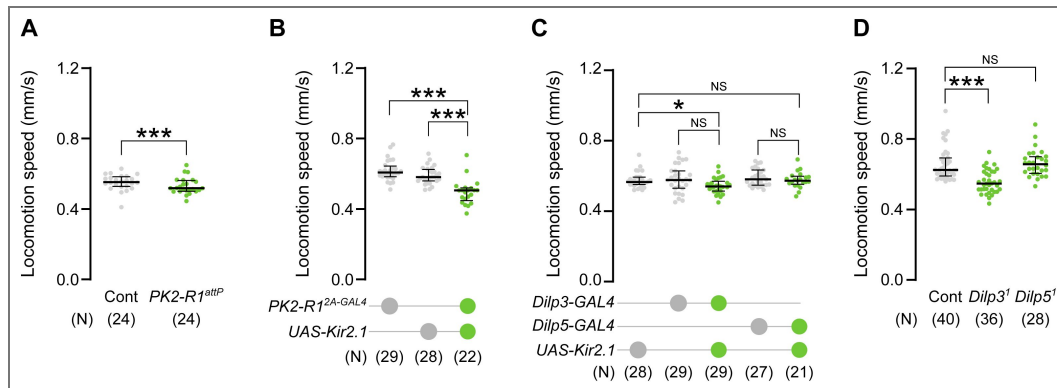
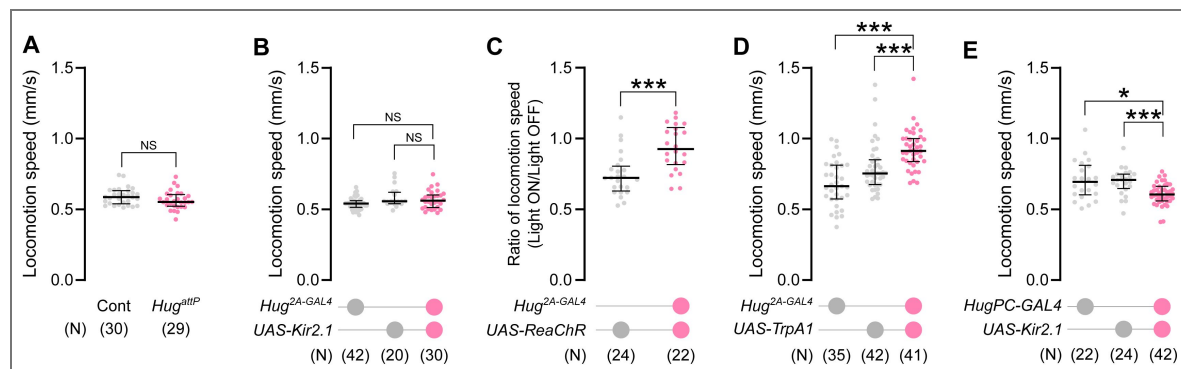


Figure 3—figure supplement 1. Effects of Hug pathway manipulations on larval locomotion speed.

(A) Locomotion speed of Hug knockout mutants during wake periods. (B) Effects of silencing Hug neurons on locomotion speed. (C-D) Effects of activating Hug neurons on locomotion speed, shown as the light ON/light OFF ratio (C) and the absolute locomotion speed (D). (E) Effects of silencing HugPC neurons on locomotion speed. Locomotion speed was quantified from the same videos used for the larval sleep assays in Figure 3. Each dot represents an individual larva; thick lines indicate the median and thin error bars indicate the interquartile range (Q1-Q3). N indicates the number of biologically independent animals per group. NS $p \geq 0.05$ (Mann-Whitney U-test) in (A). *** $p < 0.0001$, * $p < 0.05$, NS $p \geq 0.05$ (Mann-Whitney U-test with Bonferroni correction) in (B-E).



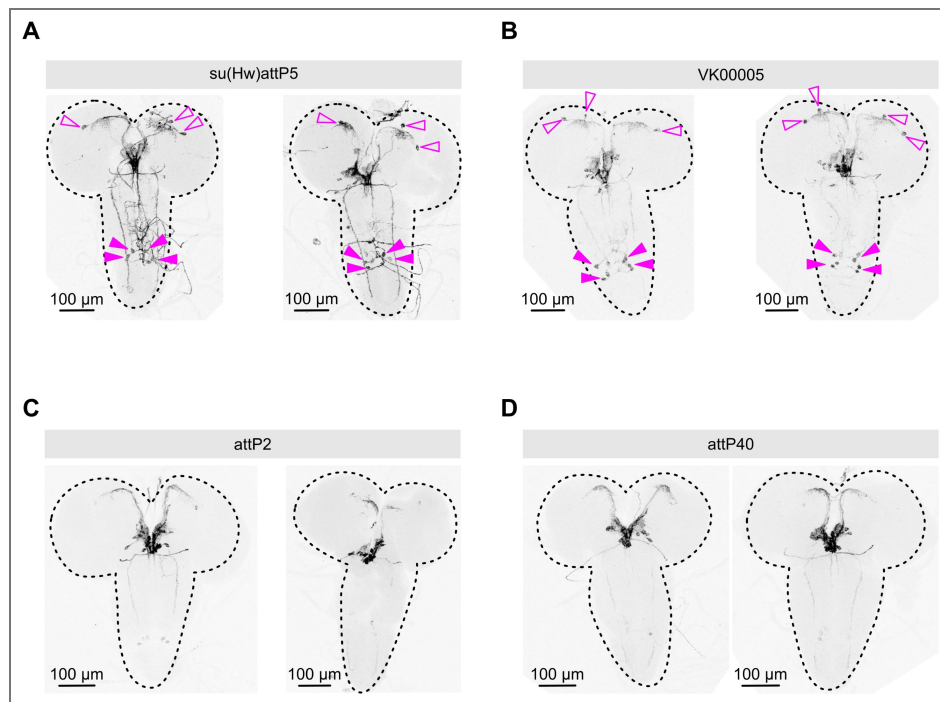


Figure 3—figure supplement 2. Generation of *HugPC-LexA* transgenic lines

(A-D) Labeling patterns of four *HugPC-LexA* strains in larval CNS visualized by rCD2::RFP. Four transgenic lines were generated by injecting the same plasmid into the following landing sites individually: su(Hw)attP5, VK00005, attP2, and attP40. Magenta arrowheads represent cell bodies apparently located outside the previously classified HugPC subpopulation (Bader et al., 2007).

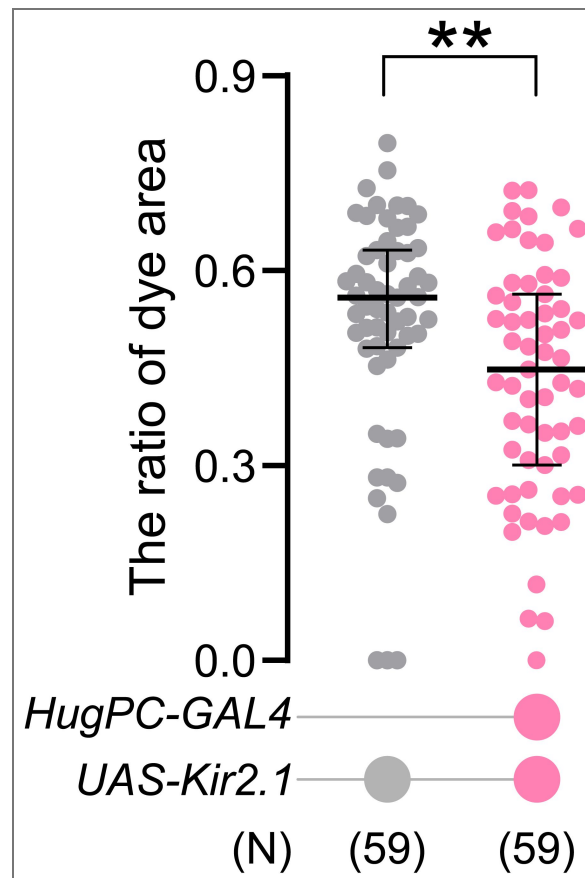


Figure 3—figure supplement 3. Silencing HugPC neurons reduces larval food intake measured by a dye-based assay.

After a 30 min starvation, larvae were allowed to feed on red yeast paste placed on an apple juice agar plate, and food intake was quantified as the ratio of the red dye-stained body surface area to the total body surface area (ImageJ). Each dot represents an individual larva; the thick line indicates the median and the thin error bars indicate the interquartile range (Q1–Q3). ** $p < 0.01$ (Mann–Whitney U-test).

Figure 4-figure supplement 1. Axonal projections of HuginPC neurons visualized with a presynaptic marker.

Representative whole-mount image of the larval central nervous system showing Syt-eGFP signals expressed in HuginPC neurons. Scale bar, 100 μ m.

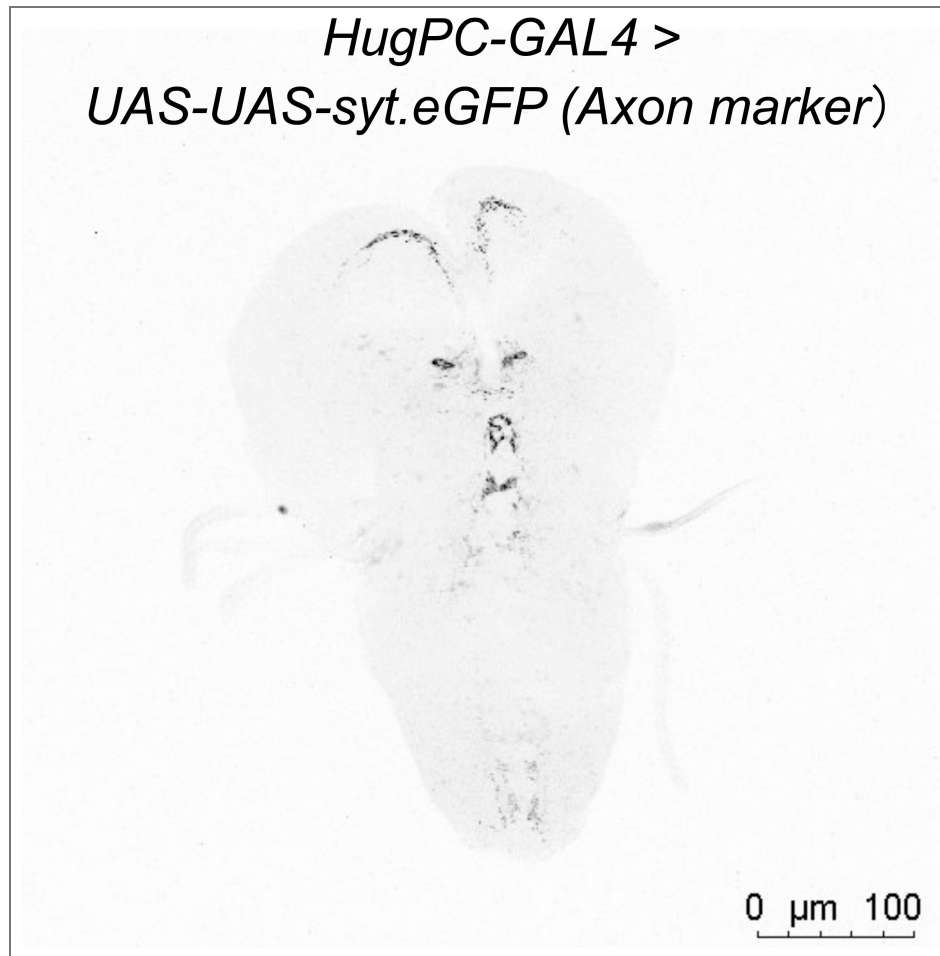
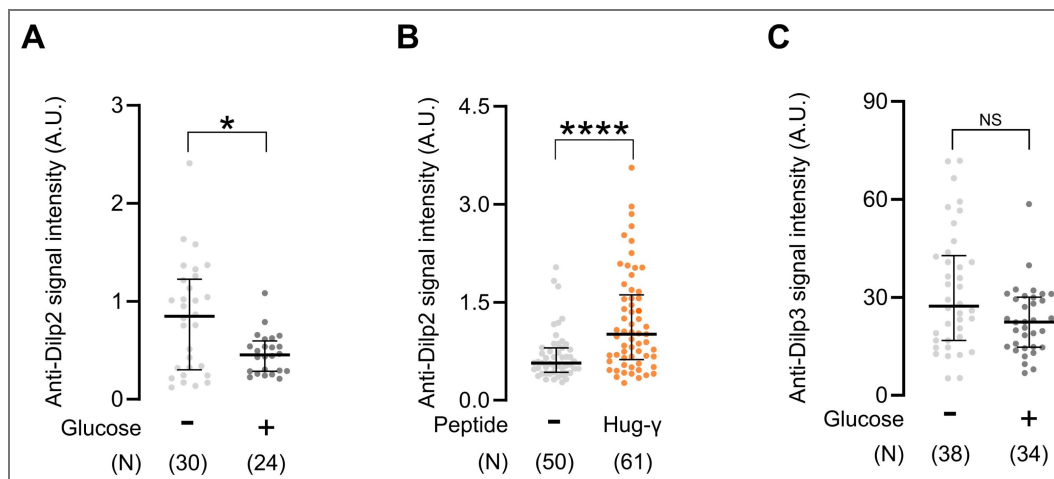


Figure 5—figure supplement 1. D-glucose but not Hug peptide cause Dilp2 reduction in larval IPCs

(A–B) Anti-Dilp2 signal intensity measured in larval IPCs after glucose (A) or Hug- γ application (B). In this and the following panels, 'N' indicates the number of cells used for each group, and the thick line and thin error bar represent the median and interquartile range (Q1–Q3), respectively. **** $p < 0.0001$, * $p < 0.05$ (Mann–Whitney U-test). (C) Anti-Dilp3 signal intensity measured within the cytosolic areas of larval IPCs. NS ≥ 0.05 (Mann–Whitney U-test).



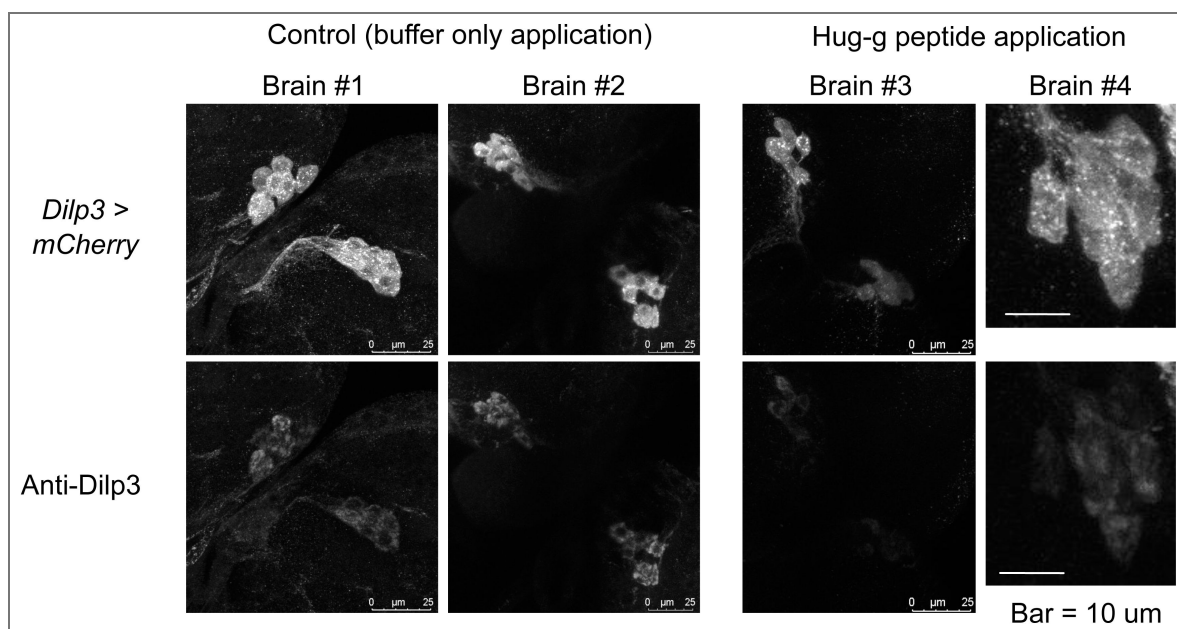


Figure 5—figure supplement 2. Representative images of Dilp3 immunoreactivity in larval IPCs after bath application of Hug-y peptide.

Larval IPCs were labeled with Dilp3>mCherry (top row), and endogenous Dilp3 was detected by anti-Dilp3 immunostaining (bottom row). Representative brains from buffer-only controls (Brain #1 and #2) and Hug-y peptide application (Brain #3 and #4) are shown. Scale bars: 25 μ m (Brains #1–3) and 10 μ m (Brain #4).

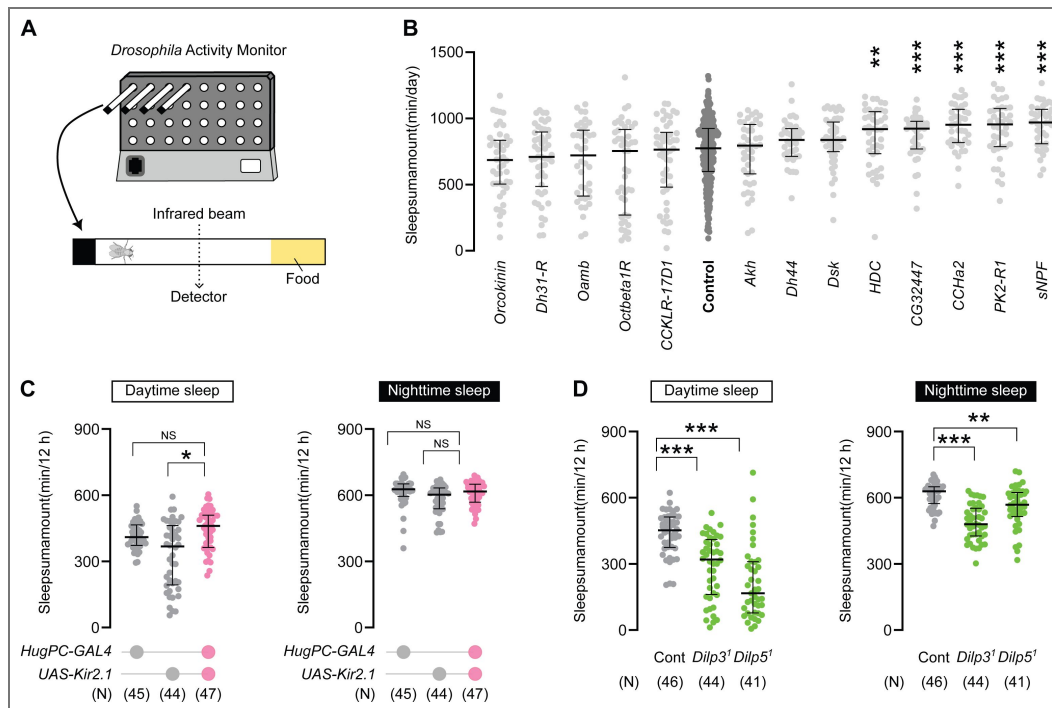


Figure 6—figure supplement 1. Sleep patterns of adult flies with genetical manipulations in the Hugin and insulin pathways

(A) Experimental setup for monitoring adult sleep using the *Drosophila* Activity Monitor (DAM). (B) Total daily sleep duration in adult flies carrying CRISPR-knockout mutations in genes whose larval sleep was significantly altered in Figure 1B. Each dot represents an individual fly; the thick line indicates the median and the thin error bars indicate the interquartile range (Q1–Q3). *** $p < 0.01$, ** $p < 0.05$ (Kruskal–Wallis one-way ANOVA followed by post hoc Mann–Whitney U-test with Bonferroni correction). (C) Daytime (left) and nighttime (right) sleep amounts in flies in which HugPC neurons were silenced. (D) Daytime (left) and nighttime (right) sleep amounts in Dilp3 or Dilp5 null mutants. In (C) and (D), numbers in parentheses indicate N, the number of biologically independent animals per group. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, NS $p \geq 0.05$ (Mann–Whitney U-test with Bonferroni correction).

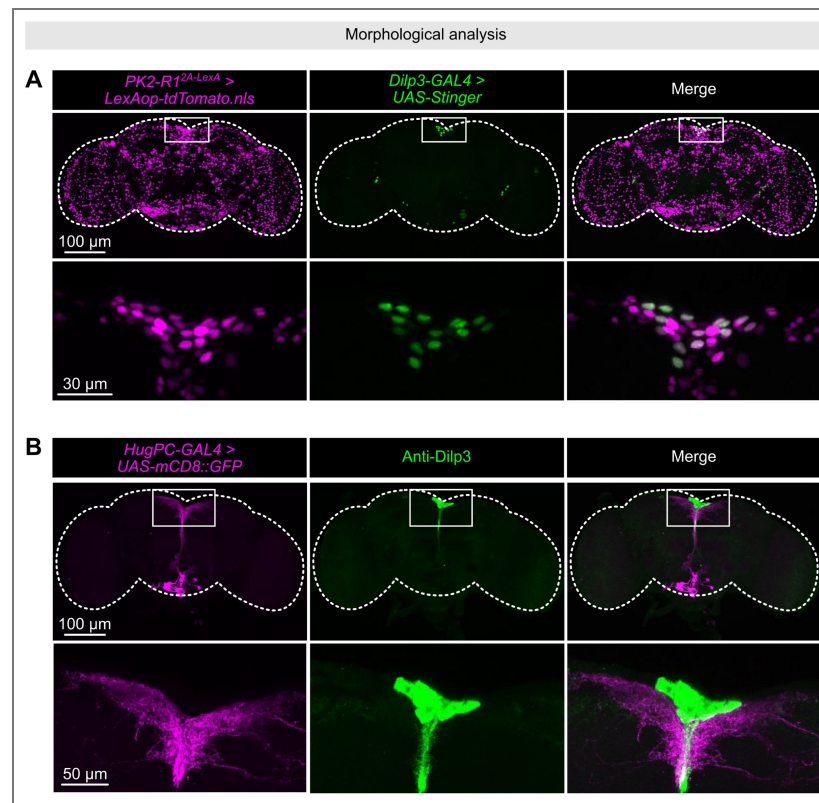


Figure 6—figure supplement 2. Morphologies of *PK2-R1* neurons, HugPC neurons, and IPCs in the adult brain.

(A) Dual labeling of *PK2-R1* neurons and IPCs expressing nuclear-localized RFP (magenta) and GFP (green), respectively. The top panels show signals in the entire adult brain. The bottom panels show magnified images of the white-squared area in each corresponding top panel, focusing on the dorsomedial brain region where the cell bodies of IPCs are located. Similar results were obtained from 5 independent samples of the same genotype. (B) Visualization of HugPC neurons and IPCs labeled by mCD8::GFP (magenta) and the anti-Dilp3 antibody (green), respectively. Similar results were obtained from 4 independent samples of the same genotype.



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Data availability

Yes, we have submitted all data sources in this article.

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Additional files

[Figure source data](#) 

Additional information

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Author ORCID iDs

Kenichi Ishii: <https://orcid.org/0000-0002-8834-5729>

References

Ahnaou A., Drinkenburg W. H. I. M (2011) Neuromedin U2 receptor signaling mediates alteration of sleep-wake architecture in rats. *Neuropeptides* **45**:165-174
<https://doi.org/10.1016/j.npep.2011.01.004>

- Bader R.**, Colomb J., Pankratz B., Schröck A., Stocker R. F., Pankratz M. J (2007) Genetic dissection of neural circuit anatomy underlying feeding behavior in *Drosophila*: Distinct classes of hugin-expressing neurons. *Journal of Comparative Neurology* **502**:848-856
<https://doi.org/10.1002/cne.21342>
- Baines R. A.**, Uhler J. P., Thompson A., Sweeney S. T., Bate M. (2001) Altered Electrical Properties in *Drosophila* Neurons Developing without Synaptic Transmission. *Journal of Neuroscience* **21**:1523-1531
- Barlow I. L.**, Rihel J. (2017) Zebrafish sleep: from geneZZZ to neuronZZZ. *Current Opinion in Neurobiology* **44**:65-71 <https://doi.org/10.1016/j.conb.2017.02.009>
- Blumberg M. S.**, H Seelke A. M., Lowen S. B., Karlsson K. A. (2005) Dynamics of sleep-wake cyclicity in developing rats. *PNAS* **102** <https://doi.org/10.1073/pnas.0506340102>
- Bonheur M.**, Swartz K. J., Metcalf M. G., Wen X., Zhukovskaya A., Mehta A., ..., Hattori D (2023) A rapid and bidirectional reporter of neural activity reveals neural correlates of social behaviors in *Drosophila*. *Nature Neuroscience* **26**:1295-1307 <https://doi.org/10.1038/s41593-023-01357-w>
- Burke C. J.**, Huetteroth W., Oswald D., Perisse E., Krashes M. J., Das G., ..., Waddell S (2012) Layered reward signalling through octopamine and dopamine in *Drosophila*. *Nature* **492**:433-437
<https://doi.org/10.1038/nature11614>
- Chiu C. N.**, Rihel J., Lee D. A., Singh C., Mosser E. A., Chen S., ..., Prober D. A (2016) A Zebrafish Genetic Screen Identifies Neuromedin U as a Regulator of Sleep/Wake States. *Neuron* **89**:842-856
<https://doi.org/10.1016/j.neuron.2016.01.007>
- Churgin M. A.**, Szuperak M., Davis K. C., Raizen D. M., Fang-Yen C., Kayser M. S (2019) Quantitative imaging of sleep behavior in *Caenorhabditis elegans* and larval *Drosophila melanogaster*. *Nature Protocols* 2019 14:5 **14**:1455-1488 <https://doi.org/10.1038/s41596-019-0146-6>
- Davis K. F.**, Parker K. P., Montgomery G. L (2004) Sleep in infants and young children: Part one: normal sleep. *Journal of Pediatric Health Care* **18**:65-71 [https://doi.org/10.1016/S0891-5245\(03\)00149-4](https://doi.org/10.1016/S0891-5245(03)00149-4)
- Deng B.**, Li Q., Liu X., Cao Y., Li B., Qian Y., ..., Rao Y (2019) Chemoconnectomics: Mapping Chemical Transmission in *Drosophila*. *Neuron* **101**:876-893.e4. <https://doi.org/10.1016/j.neuron.2019.01.045>
- Dubowy C.**, Sehgal A (2017) Circadian rhythms and sleep in *Drosophila melanogaster*. *Genetics* **205**:1373-1397 <https://doi.org/10.1534/genetics.115.185157>
- Enell L. E.**, Kapan N., Söderberg J. A. E., Kahsai L., Nässe D. R (2010) Insulin signaling, lifespan and stress resistance are modulated by metabotropic GABA receptors on insulin producing cells in the brain of *Drosophila*. *PLoS ONE* **5** <https://doi.org/10.1371/journal.pone.0015780>
- Frank M. G.**, Ruby N. F., Heller H. C., Franken P (2017) Development of circadian sleep regulation in the rat: A longitudinal study under constant conditions. *Sleep* **40** <https://doi.org/10.1093/sleep/zsw077>
- Furusawa K.**, Ishii K., Tsuji M., Tokumitsu N., Hasegawa E., Emoto K (2023) Presynaptic UBE3A E3 ligase promotes synapse elimination by downregulation of BMP signaling. *Science* **381**:1197-1205
<https://doi.org/10.1126/science.ade8978>
- Groth A. C.**, Fish M., Nusse R., Calos M. P (2004) Construction of Transgenic *Drosophila* by Using the Site-Specific Integrase From Phage ϕ C31. *Genetics* **166**:1775-1782
<https://academic.oup.com/genetics/article/166/4/1775/6052665>
- Hamada F. N.**, Rosenzweig M., Kang K., Pulver S. R., Ghezzi A., Jegla T. J., Garrity P. A (2008) An internal thermal sensor controlling temperature preference in *Drosophila*. *Nature* **454**:217-220
<https://doi.org/10.1038/nature07001>
- Hendricks J. C.**, Lu S., Kume K., Yin J. C. P., Yang Z., Sehgal A (2003) Gender dimorphism in the role of cycle (BMAL1) in rest, rest regulation, and longevity in *Drosophila melanogaster*. *Journal of Biological Rhythms* **18**:12-25 <https://doi.org/10.1177/0748730402239673>

- Hückesfeld S., Peters M., Pankratz M. J (2016) Central relay of bitter taste to the protocerebrum by peptidergic interneurons in the *Drosophila* brain. *Nature Communications* **7** <https://doi.org/10.1038/ncomms12796>
- Hückesfeld S., Schlegel P., Miroschnikow A., Schoofs A., Zinke I., Haubrich A. N., ..., Pankratz M. J (2021) Unveiling the sensory and interneuronal pathways of the neuroendocrine connectome in *Drosophila*. *eLife* <https://doi.org/10.7554/eLife.65745>
- Inagaki H. K., Jung Y., Hoopfer E. D., Wong A. M., Mishra N., Lin J. Y., ..., Anderson D. J (2014) Optogenetic control of *Drosophila* using a red-shifted channelrhodopsin reveals experience-dependent influences on courtship. *Nature Methods* **11**:325-332 <https://doi.org/10.1038/nmeth.2765>
- Lakhotia SC, Ranganath HA (2021) *Experiments with Drosophila for Biology Courses* Bengaluru, India: Indian Academy of Sciences.
- Kitatani Y., Tezuka A., Hasegawa E., Yagagi S., Togashi K., Tsuji M., Kondo S., Parrish J. Z., Emoto K (2020) *Drosophila* miR-87 promotes dendrite regeneration by targeting the transcriptional repressor Tramtrack69. *PLoS Genetics* **16**:e1008942
- Kréneisz O., Chen X., Fridell Y. W. C., Mulkey D. K (2010) Glucose increases activity and Ca²⁺ in insulin-producing cells of adult *Drosophila*. *NeuroReport* **21**:1116-1120 <https://doi.org/10.1097/WNR.0b013e3283409200>
- Kume K., Kume S., Park S. K., Hirsh J., Jackson F. R (2005) Dopamine Is a Regulator of Arousal in the Fruit Fly. *Journal of Neuroscience* **25**:7377-7384 <https://doi.org/10.1523/JNEUROSCI.2048-05.2005>
- Morikawa R., Kanamori T., Yasunaga K., Emoto K (2011) Different levels of the TRIM protein Asap regulate distinct axonal projections of *Drosophila* sensory neurons. *Proceedings of the National Academy of Sciences of the United States of America* **108**:19389-19394
- Nakamizo-Dojo M., Ishii K., Yoshino J., Tsuji M., Emoto K (2023) Descending GABAergic pathway links brain sugar-sensing to peripheral nociceptive gating in *Drosophila*. *Nature Communications* **14** <https://doi.org/10.1038/s41467-023-42202-9>
- Oh Y., Lai J. S. Y., Mills H. J., Erdjument-Bromage H., Giammarinaro B., Saadipour K., ..., Suh G. S. B (2019) A glucose-sensing neuron pair regulates insulin and glucagon in *Drosophila*. *Nature* **574**:559-564 <https://doi.org/10.1038/s41586-019-1675-4>
- Ohhara Y., Blick M., Park D., Yoon S. E., Kim Y. J., Pankratz M. J., ..., Yamanaka N (2024) A Neuropeptide Signaling Network That Regulates Developmental Timing and Systemic Growth in *Drosophila*. *The Journal of Comparative Neurology* **532**:e25677 <https://doi.org/10.1002/cne.25677>
- Okamoto N., Nishimori Y., Nishimura T (2012) Conserved role for the Dachshund protein with *Drosophila* Pax6 homolog eyeless in insulin expression. *Proceedings of the National Academy of Sciences of the United States of America* **109**:2406-2411 <https://doi.org/10.1073/pnas.1116050109>
- Omamiyuda-Ishikawa N., Sakai M., Emoto K (2020) A pair of ascending neurons in the subesophageal zone mediates aversive sensory inputs-evoked backward locomotion in *Drosophila* larvae. *PLoS Genetics* **16**:e1009120
- Shaw Paul J., Cirelli Chiara, Greenspan Ralph J., Tononi Giulio (2000) Correlates of Sleep and Waking in *Drosophila melanogaster*. *Science* **287**:1834-1837
- Poe A. R., Zhu L., Szuperak M., Mcclanahan P. D., Anafi R. C., Scholl B., ..., Kayser M. S. (2023) Developmental emergence of sleep rhythms enables long-term memory in *Drosophila*. <https://www.science.org>
- Raizen D. M., Zimmerman J. E., Maycock M. H., Ta U. D., You Y. J., Sundaram M. V., Pack A. I (2008) Lethargus is a *Caenorhabditis elegans* sleep-like state. *Nature* **451**:569-572 <https://doi.org/10.1038/nature06535>
- Rosenkilde C., Cazzamali G., Williamson M., Hauser F., Søndergaard L., DeLotto R., Grimmelikhuijzen C. J. P (2003) Molecular cloning, functional expression, and gene silencing of two *Drosophila* receptors for the *Drosophila* neuropeptide pyrokinin-2. *Biochemical and Biophysical Research Communications* **309**:485-494 <https://doi.org/10.1016/j.bbrc.2003.08.022>

- Scammell T. E., Arrigoni E., Lipton J. O. (2017) Neural Circuitry of Wakefulness and Sleep. *Neuron* **93**:747-765 <https://doi.org/10.1016/j.neuron.2017.01.014>
- Schlegel P., Texada M. J., Miroshnikow A., Schoofs A., Hückesfeld S., Peters M., ..., Pankratz M. J. (2016) Synaptic transmission parallels neuromodulation in a central food-intake circuit. *eLife* **5**:e16799 <https://doi.org/10.7554/eLife.16799>
- Schneider C. A., Rasband W. S., Eliceiri K. W. (2012) NIH Image to ImageJ: 25 years of image analysis. *Nature Methods* **9**:671-675 <https://doi.org/10.1038/nmeth.2089>
- Schoofs A., Hückesfeld S., Schlegel P., Miroshnikow A., Peters M., Zeymer M., ..., Pankratz M. J. (2014) Selection of Motor Programs for Suppressing Food Intake and Inducing Locomotion in the *Drosophila* Brain. *PLoS Biology* **12** <https://doi.org/10.1371/journal.pbio.1001893>
- Schwarz J. E., King A. N., Hsu C. T., Barber A. F., Sehgal A. (2021) Hugin + neurons provide a link between sleep homeostat and circadian clock neurons. *PNAS* <https://doi.org/10.1073/pnas.2111183118/DCSupplemental>
- Shafer O. T., Keene A. C. (2021) The Regulation of *Drosophila* Sleep. *Current Biology* **31**:R38-R49 <https://doi.org/10.1016/j.cub.2020.10.082>
- Sorribes A., Porsteinsson H., Arnardóttir H., Jóhannesdóttir I. P., Sigurgeirsson B., de Polavieja G. G., Karlsson K. (2013) The ontogeny of sleep-wake cycles in zebrafish: A comparison to humans. *Frontiers in Neural Circuits* **7** <https://doi.org/10.3389/fncir.2013.00178>
- Szuperak M., Churgin M. A., Borja A. J., Raizen D. M., Fang-Yen C., Kayser M. S. (2018) A sleep state in *Drosophila* larvae required for neural stem cell proliferation. *eLife* **7** <https://doi.org/10.7554/eLife.33220>
- Veenstra J. A., Agricola H. J., Sellami A. (2008) Regulatory peptides in fruit fly midgut. *Cell and Tissue Research* **334**:499-516 <https://doi.org/10.1007/s00441-008-0708-3>
- Yoshino J., Morikawa R., Hasegawa E., Emoto K. (2017) Neural circuitry that evokes escape behavior in response to nociceptive stimuli in *Drosophila* larvae. *Current Biology* **27**:2499-2504
- Yoshino J., Mali S., Williams C.R., Morita T., Emerson C., Arp C., Miller S., Yin C., Motoyoshi M., Hemmi C., et al. (2025) *Drosophila* epidermal cells are intrinsically mechanosensitive and modulate nociceptive behavior outputs. *eLife* **13**:R95379 <https://doi.org/10.7554/eLife.95379>

Peer reviews

Reviewer #1 (Public review):

The manuscript investigates how neuropeptidergic signaling affects sleep regulation in *Drosophila* larvae. The authors first conduct a screen of CRISPR knock-out lines of genes encoding enzymes or receptors for neuropeptides and monoamines. As a result of this screen, the authors follow up on one hit, the hugin receptor, PK2-R1. They use genetic approaches including mutants and targeted manipulations of PK2-R1 activity in insulin-producing cells (IPCs) to increase total sleep amounts in 2nd instar larvae. Similarly, *dilp3* and *dilp5* null mutants and genetic silencing of IPCs show increases in sleep. The authors also show that hugin mutants and thermogenetic/optogenetic activation of hugin-expressing neurons caused reductions in sleep. Furthermore, they show through imaging-based approaches that hugin-expressing neurons activate IPCs. A key finding is that wash on of hugin peptides, Hug-γ and PK-2, in ex vivo brain preparations activates larval IPCs, as assayed by CRTC::GFP imaging. The authors then examine how the PK2-R1, hugin, and IPC manipulations affect adult sleep. Finally, the authors examine how Ca²⁺ responses through CRTC::GFP imaging in adult IPCs are influenced by the wash on of hugin peptides.

Strengths:

- (1) This paper builds on previously published studies that examine *Drosophila* larval sleep regulation. Through the power of *Drosophila* genetics, this study yields additional insights

into what role neuropeptides play in regulation of *Drosophila* larval sleep.

(2) This study utilizes several diverse approaches to examine larval and adult sleep regulation, neural activity, and circuit connections. The impressive array of distinct analyses provides new understanding into how *Drosophila* sleep-wake circuitry is regulated across the lifespan.

(3) The imaging approaches used to examine IPC activation upon hugin manipulation (either thermogenetic activation or wash on of peptides) demonstrate a powerful approach for examining how changes in neuropeptidergic signaling affect downstream neurons. These experiments involve precise manipulations as the authors use both *in vivo* and *ex vivo* conditions to observe an effect on IPC activity.

Weaknesses:

(1) There is limited discussion of why statistically significant differences are observed in some genetic and temperature controls. This discussion would better support the authors' conclusions.

(2) The functional connectivity of the huginPC-IPC circuit in larvae could be better supported by chemogenetics using real-time calcium imaging (GCaMP).

Comments on revisions:

I would like to thank the authors for the revisions. The inclusion of all sleep metrics, more detailed descriptions in the methods, & a more thorough comparison to other published articles has addressed most of my concerns.

<https://doi.org/10.7554/eLife.105710.2.sa2>

Reviewer #3 (Public review):

Summary:

Sleep affects cognition and metabolism, evolving throughout development. In mammals, infants have fast sleep-wake cycles that stabilize in adults via circadian regulation. In this study, the author performed a genetic screen for neurotransmitters/peptides regulating sleep and identified the neuropeptide Hugin and its receptor PK2-R1 as essential components for sleep in *Drosophila* larvae. They showed that IPCs express Pk2-R1 and silencing IPCs resulted in significant increase in the sleep amount, which was consistent with the effect they observed in PK2-R1 knock out mutants. They also showed that Hugin peptides, secreted by a subset of Hugin neurons (Hug-PC), activate IPCs through the PK2-R1 receptor. This activation prompts IPCs to release insulin-like peptides (Dilps), which are implicated in the modulation of sleep. They showed that Hugin peptides induce a PK2-R1 dependent calcium (Ca^{2+}) increase in IPCs, which they linked to the release of Dilp3, showing a connection between Hugin signaling to IPCs, Dilp3 release and sleep regulation. Additionally, the activation of Hug-PC neurons reduced sleep amounts, while silencing them had the opposite effect. In contrast to the larval stage, the Hugin/PK2-R1 axis was not critical for sleep regulation in *Drosophila* adults, suggesting that this neuropeptidergic circuitry has divergent roles in sleep regulation across different stages of development.

Strengths:

This study used an updated system for sleep quantification in *Drosophila* larvae and this method allowed precise measurement of larval sleep patterns which is essential for the understanding of sleep regulation.

The authors performed unbiased genetics screening and successfully identified novel regulators for larval sleep, Hugin and its receptor PK2-R1, making a substantial contribution to the understanding of neuropeptidergic control of sleep regulation.

They clearly demonstrated the mechanism by which Hugin expressing neurons influence sleep through the activation of IPCs via PK2-R1 with Ca²⁺ responses and can modulate sleep.

Based on the demonstrated activation of PK2-R1 by the human Hugin orthologue Neuromedin U, research on human sleep disorders may benefit from the discoveries from *Drosophila* since sleep regulating mechanisms are conserved across species.

Weaknesses:

Previously identified weaknesses have been largely addressed by the authors.

<https://doi.org/10.7554/eLife.105710.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

*The study investigates how neuropeptidergic signaling affects sleep regulation in *Drosophila* larvae. The authors first conduct a screen of CRISPR knock-out lines of genes encoding enzymes or receptors for neuropeptides and monoamines. As a result of this screen, the authors follow up on one hit, the hugin receptor, PK2-R1. They use genetic approaches, including mutants and targeted manipulations of PK2-R1 activity in insulin-producing cells (IPCs) to increase total sleep amounts in 2nd instar larvae. Similarly, *dilp3* and *dilp5* null mutants and genetic silencing of IPCs show increases in sleep. The authors also show that hugin mutants and thermogenetic/optogenetic activation of hugin-expressing neurons caused reductions in sleep. Furthermore, they show through imaging-based approaches that hugin-expressing neurons activate IPCs. A key finding is that wash-on of hugin peptides, Hug-y and PK-2, in ex vivo brain preparations activates larval IPCs, as assayed by CRTC::GFP imaging. The authors then examine how the PK2-R1, hugin, and IPC manipulations affect adult sleep. Finally, the authors examine how Ca²⁺ responses through CRTC::GFP imaging in adult IPCs are influenced by the wash-on of hugin peptides. The conclusions of this paper are somewhat well supported by data, but some aspects of the experimental approach and sleep analysis need to be clarified and extended.*

Strengths:

*(1) This paper builds on previously published studies that examine *Drosophila* larval sleep regulation. Through the power of *Drosophila* genetics, this study yields additional insights into what role neuropeptides play in the regulation of *Drosophila* larval sleep.*

*(2) This study utilizes several diverse approaches to examine larval and adult sleep regulation, neural activity, and circuit connections. The impressive array of distinct analyses provides new understanding into how *Drosophila* sleep-wake circuitry is regulated across the lifespan.*

(3) The imaging approaches used to examine IPC activation upon hugin manipulation (either thermogenetic activation or wash-on of peptides) demonstrate a powerful

approach for examining how changes in neuropeptidergic signaling affect downstream neurons. These experiments involve precise manipulations as the authors use both *in vivo* and *ex vivo* conditions to observe an effect on IPC activity.

Weaknesses:

Although the paper does have some strengths in principle, these strengths are not fully supported by the experimental approaches used by the authors. In particular:

- (1) The authors show total sleep amount over an 18-hour period for all the measures of 2nd instar larval sleep throughout the paper. However, published studies have shown that sleep changes over the course of 2nd instar development, so more precise time windows are necessary for the analyses in this study.
- (2) Previously published reports of sleep metrics in both *Drosophila* larvae and adults include the average number of sleep episodes (bout number) and the average length of sleep episodes (bout length). Neither of these metrics is included in the paper for either the larval sleep or adult sleep data. Not including these metrics makes it difficult for readers to compare the findings in this study to previously published papers in the established *Drosophila* sleep literature.
- (3) Because *Drosophila* adult & larval sleep is based on locomotion, the authors need to show the activity values for the experiments supporting their key conclusions. They do show travel distances in Figure 2 - Figure Supplement 1, however, it is not clear how these distances were calculated or how the distances relate to the overall activity of individual larvae during sleep experiments. It is also concerning that inactivation of the PK2-R1-expressing neurons causes a reduction in locomotion speed. This could partially explain the increase in sleep that they observe.
- (4) The authors rely on homozygous mutant larvae and adult flies to support many of their conclusions. They also rely on Gal4 lines with fairly broad expression in the *Drosophila* brain to support their conclusions. Adding more precise tissue-specific manipulations, including thermogenetic activation and inhibition of smaller populations of neurons in the study would be needed to increase confidence in the presented results. Similarly, demonstrating that larval development and feeding are not affected by the broad manipulations would strengthen the conclusions.
- (5) Many of the experiments presented in this study would benefit from genetic and temperature controls. These controls would increase confidence in the presented results.
- (6) The authors claim that their findings in larvae uncover the circuit basis for larval sleep regulation. However, there is very little comparison to published studies demonstrating that neuropeptides like Dh44 regulate larval sleep. Because hugin-expressing neurons have been shown to be downstream of Dh44 neurons, the authors need to include this as part of their discussion. The authors also do not explain why other neuropeptides in the initial screen are not pursued in the study. Given the effect that these manipulations have on larval sleep in their initial screen, it seems likely that other neuropeptidergic circuits regulate larval sleep.

We thank Reviewer #1 for the constructive comments. According to the suggestions, we have compared the relative sleep amounts of wild-type control and Hugin/PK2-R1/IPC mutants/manipulations between 6hr-period and 18-hour periods in the 2nd instar larval stage and found consistent sleep phenotypes. We have also showed the sleep metrics data of larva and adults. We have included additional data of locomotion and feeding behavior in wild-type control and Hugin/PK2-R1/IPC mutants/manipulations, which suggest that sleep phenotypes of Hugin/PK2-R1/IPC mutants/manipulations are less affected by locomotion and feeding behavior changes. As pointed out, our study could not exclude the possibility that in

addition to the Hugin/PK2-R1/IPC axis, other pathways including DH44 could act in larval sleep control. We have included these points in Discussion. Please see point-to-point responses for details.

Reviewer #2 (Public review):

Summary:

This study examines larval sleep patterns and compares them to sleep regulation in adult flies. The authors demonstrate hallmark sleep characteristics in larvae, including sleep rebound and increased arousal thresholds. Through genetic and behavioral analyses, they identify PK2-R1 as a key receptor involved in sleep modulation, likely via the HuginPC-IPC signaling pathway. Loss of PK2-R1 results in increased sleep, which aligns with previous findings in hugin knockout mutants. While the study presents significant contributions to the field, further investigation is needed to address discrepancies with earlier research and strengthen mechanistic claims.

Strengths:

(1) The study explores a relatively understudied aspect of sleep regulation, focusing on larval development.

(2) The use of an automated behavioral measurement system ensures precise quantification of sleep patterns.

(3) The findings provide strong genetic and behavioral evidence supporting the role of the HuginPC-IPC pathway in sleep regulation.

(4) The study has broader implications for understanding the evolution and functional divergence of sleep circuits.

Weaknesses:

(1) The manuscript does not sufficiently discuss previous studies, particularly concerning hugin mutants and their metabolic effects.

(2) The specificity of IPC secretion mechanisms is unclear, particularly regarding potential indirect effects on Dilp2.

(3) Alternative circuits, such as the HuginPC-DH44 pathway, require further consideration.

(4) Functional connectivity between HuginPC neurons and IPCs is not directly validated.

(5) Developmental differences in sleep regulatory mechanisms are not thoroughly examined.

We thank Reviewer #2 for the positive comments. As suggested, our study could not exclude the possibility that in addition to the Hugin/PK2-R1/IPC axis, alternative pathways including the Hugin/DH44 axis could contribute to sleep control in larvae. We have added these points in Discussion. We also have added additional data to show mechanistic differences of larval and adult sleep control. Please see point-to-point responses for details.

Reviewer #3 (Public review):

Summary:

Sleep affects cognition and metabolism, evolving throughout development. In mammals, infants have fast sleep-wake cycles that stabilize in adults via circadian regulation. In this study, the author performed a genetic screen for neurotransmitters/peptides regulating

sleep and identified the neuropeptide Hugin and its receptor PK2-R1 as essential components for sleep in *Drosophila* larvae. They showed that IPCs express Pk2-R1 and silencing IPCs resulted in a significant increase in the sleep amount, which was consistent with the effect they observed in PK2-R1 knock-out mutants. They also showed that Hugin peptides, secreted by a subset of Hugin neurons (Hug-PC), activate IPCs through the PK2-R1 receptor. This activation prompts IPCs to release insulin-like peptides (Dilps), which are implicated in the modulation of sleep. They showed that Hugin peptides induce a PK2-R1 dependent calcium (Ca^{2+}) increase in IPCs, which they linked to the release of Dilp3, showing a connection between Hugin signaling to IPCs, Dilp3 release, and sleep regulation. Additionally, the activation of Hug-PC neurons reduced sleep amounts, while silencing them had the opposite effect. In contrast to the larval stage, the Hugin/PK2-R1 axis was not critical for sleep regulation in *Drosophila* adults, suggesting that this neuropeptidergic circuitry has divergent roles in sleep regulation across different stages of development.

Strengths:

*This study used an updated system for sleep quantification in *Drosophila* larvae, and this method allowed precise measurement of larval sleep patterns which is essential for the understanding of sleep regulation.*

The authors performed unbiased genetics screening and successfully identified novel regulators for larval sleep, Hugin and its receptor PK2-R1, making a substantial contribution to the understanding of neuropeptidergic control of sleep regulation.

They clearly demonstrated the mechanism by which Hugin-expressing neurons influence sleep through the activation of IPCs via PK2-R1 with Ca^{2+} responses and can modulate sleep.

*Based on the demonstrated activation of PK2-R1 by the human Hugin orthologue Neuromedin U, research on human sleep disorders may benefit from the discoveries from *Drosophila* since sleep-regulating mechanisms are conserved across species.*

Weaknesses:

*The study primarily focused on sleep regulation in *Drosophila* larvae, showing that the Hugin/PK2-R1 axis is critical for larval sleep but not necessary for adult sleep. The effects of the Hugin axis in the adult are, however, incompletely explained and somewhat inconsistent. PK2-R1 knockout adults also display increased sleep, as does HugPC silencing, at least for daytime sleep. The difference lies in Dilp3/5 mutant animals showing decreased sleep and IPCs seemingly responding with reduced Dilp3 release to PK-2 treatment (Figure 6). It seems difficult to reconcile the author's conclusions regarding this point without additional data. It could be argued that PK2-R1 still regulates adult sleep, but not via Hugin and IPCs/Dilps.*

Another issue might be that the authors show relative sleep levels for adults using Trikinetics monitoring. From the methods, it is not clear if the authors backcrossed their line to an isogenic wild-type background to normalize for line-specific effects on sleep. Thus, it is likely that each line has differences in total sleep time due to background effects, e.g., their Kir2.1 control line showed reduced sleep relative to the compared genotypes. This might limit the conclusions on the role of Hugin/PK2-R1 on adult sleep.

We thank Reviewer #3 for the valuable comments. According to the suggestions, we have included additional data of adult sleep phenotypes with IPCs/Dilps and HugPC/PK-2 manipulations. We believe that these additional data further support the idea that the Hugin/PR2/IPC axis acts differently in larval and adult sleep control.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) Show all data as individual data points in the graphs. The use of box-and-whisker plots makes it difficult to determine how much variation there is in each experiment.

According to the comments, we have changed all graphs to the dots-and-whisker plots (Figures 1–6; Figure 1—figure supplements 2–4; Figure 2—figure supplement 1; Figure 3—figure supplement 1 and 3; Figure 5—figure supplement 1; and Figure 6—figure supplements 1 and 3).

(2) Show all larval sleep metrics (total sleep duration, bout #, bout length, & activity) over the first 6-hour period of 2nd instar development. Larval sleep changes over the course of 2nd instar development so showing an 18-hour period is not as informative for the different manipulations in the study. This also allows for a more thorough comparison to Szuperak et al 2018.

According to the comments, we have shown all larval sleep metrics (total sleep duration, bout #, bout length, & activity) over the first 6 hours for PK2-R1 KO mutants (Figure 1-figure supplemental 5). These PK2-R1 mutant phenotypes are consistent with those described by our sleep amount data over an 18 hr period (Figure 1-figure supplemental 5). We thus consistently show all the sleep phenotype data in the 18 hr period window in the 2nd instar larvae in this paper.

(3) Show activity values for every experiment. Behavior is based on locomotion, so there is a need to show that larvae in each manipulation do not have locomotive defects.

According to the reviewer's comments, we have shown the activity values for each experiment (Figure 2—figure supplement 1 and Figure 3—figure supplement 1). These data clearly indicated that changes in sleep amounts in each manipulation are not only due to locomotion alterations. We have thus added the sentence below at line 151156 in the manuscript.

Locomotion changes were not consistently observed upon either activation or suppression of Hug neurons (Figure 3—figure supplement 1), suggesting that changes in sleep amounts is unrelated to locomotor alterations.

(4) Provide additional explanation as to why PK2-R1 was pursued in the study. There are several candidates in Figure 1 - Figure Supplement 4 (like sNPF-Gal4, Dh31-Gal4, and DskGal4) that have effects on sleep. These have also not been studied in the context of larval sleep regulation.

According to the reviewer's comments, we have added the following sentences at line 108-114 in the manuscript.

The role of PK2-R1 in larval sleep, on the other hand, has been unknown to date. Given its strong expression in insulin-producing cells (Schlegel et al., 2016) and its function as a receptor for the neuropeptide Hugin, which modulates feeding (Schoofs et al., 2014), we hypothesized that PK2-R1 might mediate neuropeptidergic signaling that links metabolic and sleep regulation during development. We thus focused on this gene as a candidate connecting behavioral and endocrine sleep control.

(5) Insulin manipulations are known to disrupt Drosophila development (Rulifson et al, 2002). Therefore, it would be beneficial to show that larvae develop normally in dilp3 and dilp5 mutants by examining the time to pupal formation in these mutants compared to controls. If the mutant larvae take longer to reach the pupal stage, how do the authors know that the 2nd instar control and mutant larvae are the same developmental age? As indicated above, the developmental age of larvae does affect the total amount of sleep, so this could affect the authors' conclusions.

We agree that this is an important point in this study. In each experiment, we carefully checked the developmental stage of larvae progeny by mouth hook analysis and measuring larval size and used only larvae with characteristics comparable to wildtype 2nd instar larvae. We have added these descriptions in Methods (line 411–416).

(6) Figure 1 data is only supported by homozygous mutants & 1 fairly-broadly expressed Gal4 driver. The authors need to show that inactivation of PK2-R1 neurons with more tissue-restrictive Gal4 driver lines has the same effect as the other manipulations to further support the conclusions. Examining sleep in activation of PK2-R1 neurons with the broadly expressed Gal4 driver & UAS-TrpA1 would also provide better support for the conclusions.

We agree. Indeed, we tried to narrow down to small subsets of neurons using multiple different Gal4 drivers, but unfortunately, we did not obtain potential candidates.

Therefore, although our data show that the Hugin/PK2-R1 axis contributes to sleep control in larvae, we cannot rule out the possibility that other axes could also function in larval sleep control. We mentioned this point in the original version of the submitted manuscript (line 134–137).

(7) Provide more explanation as to how your methods of defining sleep compare/contrast to published papers. It is not clear how many frames = 1 sec in your recordings. The definition of sleep as 12 frames needs to include a time component as well. This allows for easier comparison to other published papers examining Drosophila larval sleep (Szuperak et al 2018; Churgin et al 2019; Poe et al 2023; Poe et al 2024).

Our recordings were acquired at 0.87 frames per second. We have added this information in Method (line 431).

(8) Figure 2 data is only supported by mutants & inactivation with 1 Gal4 driver per cell population. Showing activation of Gal4-expressing cells with UAS-TrpA1 would add more support to the conclusions.

We have already showed the reduced sleep amounts in both HuginGAL4>ReaChR and HuginGAL4>TrpA larvae (Figure 3 C & D) in the original version.

(9) Need to clarify in the methods how the authors calculated travel distances as a measure of locomotive activity. It's not clear if this is done during larval sleep experiments or in independent experiments. It is also not clear why the y-axes of Figure 2—Figure Supplement 1 are not consistent across the panels. Finally, the authors do see decreases in locomotive activity in PK2-R1>Kir2.1 and in dilp3 mutants, so the conclusions presented in the results section of the paper need to be modified to reflect those results.

We calculated travel distances from the same video recording datasets used for sleep quantification. We have added this information in Method (line 431–435). As the reviewer indicated, locomotor activity was reduced in a part of conditions/mutants including PK2-R1 > Kir2.1 and dilp3 mutants, and therefore we cannot exclude the possibility that locomotion changes might contribute to sleep phenotypes. On the other hand, a large part of manipulations of Hugin neurons and IPCs caused a sleep increase without significant changes in locomotor activity (Figure 2—figure supplement 1 and Figure 3—figure supplement 1). It is thus likely that Hugin and IPCs contribute to sleep control independent of locomotion, whereas other neurons trapped by PK2-R1 GAL4 might contribute to locomotion control.

(10) Given the role that hugin neurons play in Drosophila feeding (Schlegel et al, 2016), the authors should include feeding data for the hugin/PK2-R1 manipulations. It is also unclear from the methods if their thresholding for defining sleep can detect feeding behaviors. Changes in feeding behavior could explain some of the reported increases in sleep if feeding is not classified as a waking but is instead picked up as inactivity.

We agree that this is an important point. According to reviewer's points, we have added feeding amounts of the wild-type control and the HuginPC>Kir2.1 larvae (Figure 3—figure supplement 3). These data suggest that feeding amounts of the HuginPC>Kir2.1 larvae are significantly reduced compared to those of the control. Given that our data analysis typically categorized feeding behavior into “moving (not sleep)” (see Materials and Methods) and that HuginPC>Kir2.1 larvae showed increased sleep amounts compared to the wild-type control, it is likely that the increased sleep amounts in HuginPC>Kir2.1 larvae are unrelated to changes in feeding behavior.

(11) The Hugin-IPC localization data (Figure 3E) would be better supported by the use of more specific synaptic and dendritic markers. Specifically, expressing Syt-eGFP (axon marker) in hugin neurons & DenMark (dendritic marker) in IPCs. Using GRASP or P2X2 to demonstrate the anatomical/functional connections between hugin & IPC neurons would also provide better support for this conclusion.

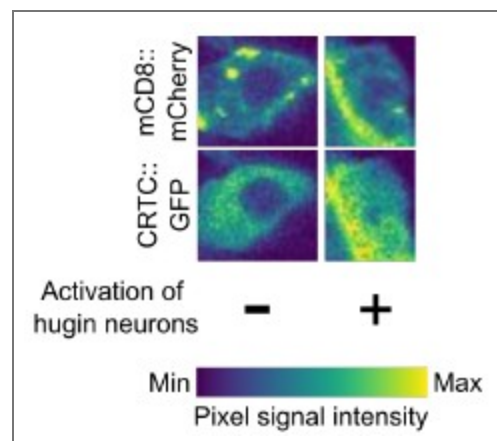
According to the reviewer's suggestion, we have added Syt-eGFP signals in HuginPC neurons (Figure 4—figure supplement 1). We also tried DenMark expression in IPCs, but we could not obtain *dipl3>DenMark* F1 progeny for unknown reason. We also applied GRASP to the HuginPC-IPC interaction, but we could not detect obvious GRASP signals. It is well known that peptidergic transmission is often independent of conventional synapse structures, called as volume transmission, in which peptidergic signals can transmit over a long-range distance to targeting neurons. It is thus possible that IPCs might receive Hugin signals from HuginPC neurons through volume transmission.

(12) Figure 4 is missing temperature controls for thermal activation experiments. Also missing genetic control for UAS/+. It would be more convincing to see experiments in Figure 4 with the more specific hug-PC-Gal4 line instead of the broadly expressed hugin-Gal4 line.

According to reviewer's comments, we have added the control data in Figure 4.

(13) Representative images for Figure 4B & 4C would provide better support for the quantifications & conclusions presented.

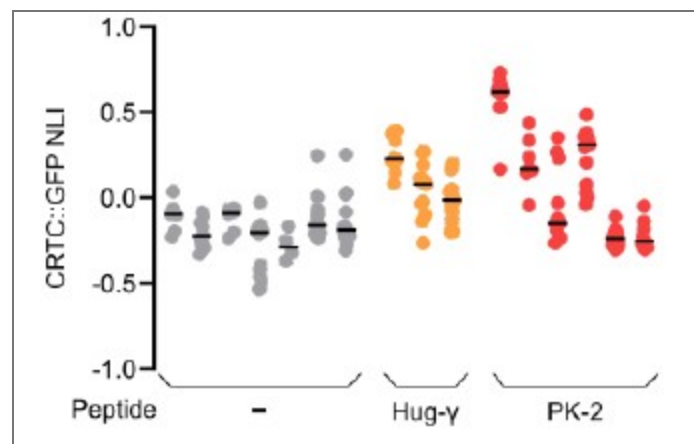
According to the reviewer's suggestions, we show the representative image for Figure 4B and 4C (please see Author response image 1). We are, however, afraid that these images might not help readers' further understanding in addition to the quantitative data, so we have decided to not add these images in the manuscript.



Author response image 1. mCD8::mCherry (top) and CRTC::GFP (bottom) are shown under high-temperature conditions without ("-") or with ("+") hugin neuron activation. "-" denotes a high-temperature genetic control lacking LexAop-TrpA1, thus no thermogenetic activation occurs. CRTC::GFP is shown in pseudocolor.

(14) A more zoomed-out image of all the IPC neurons in the bath application of hugin peptides (Figure 5D) would help with the interpretation of the results. It's not clear if the authors only measured the same exact neuron in each IPC cluster or if they examined all of the IPC neurons. If they measured all of the IPC neurons, did they observe similar results across the different neurons? How much variability is there in the response of IPC neurons to hugin peptide application?

For Figure 5, we obtained images of multiple brains from each genotype and quantified the NLI values from all IPC neurons. For reference, we show plots of the CRTC signals of Figure 5C each brain by brain (Author response image 2). We have added detailed information of CRTC analysis in Methods (lines 552-554).



Author response image 2. Distribution of CRTC signals across individual brains. Plots of nuclear localization index (NLI) for individual brains, corresponding to the conditions shown in Figure 5C. The x-axis represents each larval brain preparation, and each dot indicates the NLI value of a single IPC neuron. Horizontal bars represent the median within each brain. These plots illustrate variability both within and across individual brains.

(15) The conclusion that application of Hug peptides results in dilp3 release is not well supported (Figure 5E). There is a large amount of variation in anti-dilp3 signal. Representative images for these quantifications would be beneficial. The authors also don't directly show that dilp3 vesicles are released. They only see a reduction in antibody accumulation in IPCs. Could there be other reasons for the reduction in accumulation in the IPCs? Would changes in dilp3 gene expression or membrane localization cause a

reduction in signal? Showing that actual release of dilp3 is affected by Hug peptides using a reporter like ANF-GFP would be more convincing.

According to the reviewer's comments, we have added representative images (Figure 5—figure supplement 2). As for the ex vivo experiments in Fig5, we treated the extracted brain tissues with Hugin/NMU peptides for only 5minutes. It is thus most likely that reduction of Dilps in IPCs is mediated by Hugin/PK2-R1 signal-dependent secretion, rather than transcriptional control and/or degradation of Dilps.

(16) Show all sleep metrics (total sleep duration, bout #, bout length, and activity) for adult sleep experiments. Showing relative total sleep for the adult experiments is confusing & would benefit from plots of total average sleep in minutes for each genotype.

According to the reviewer's comments, we have added the sleep metrics in adults (Figure 6; Figure 6-figure supplement 3).

(17) The authors can't conclude that expression patterns of PK2-R1 & hug between larvae & adults are "almost comparable." They don't track neurons over development or immortalize neurons in larvae & check expression patterns in adults. They need to show some type of quantification to support these claims. Or revise the text to remove this conclusion.

We agree. We have changed our arguments as follow (line 211-214).

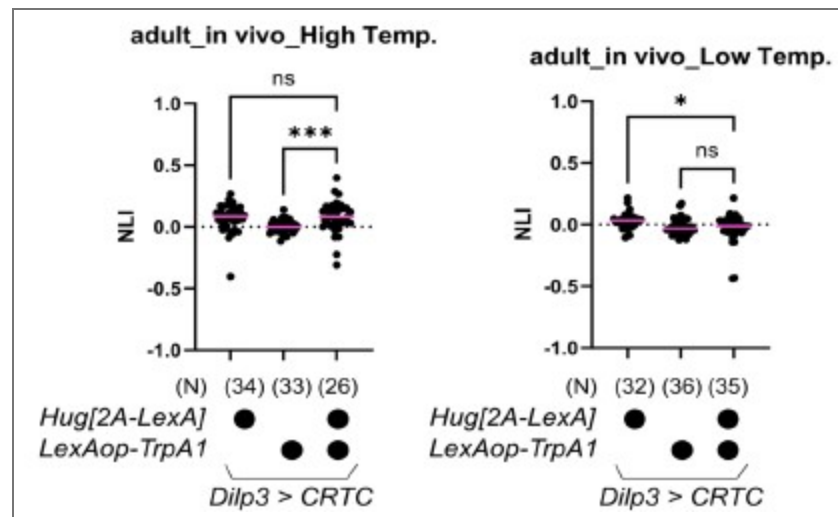
Interestingly, the expression patterns of PK2-R1 and Hug as well as the morphology of HugPC neurons in adults appeared to be similar to those in larvae (Figure 6—figure supplement 2), implying that the differential roles of Hug in larvae vs adults are likely due to physiological differences in HugPC neurons and/or IPCs.

(18) For Figure 6, what effect does genetic inactivation of IPCs have on adult sleep? A more specific manipulation of these cells would provide better support for the conclusion that IPC manipulations have distinct effects on larval & adult sleep. The sleep traces for the hugin manipulation & dilp mutants (Figure 6-Figure Supplement 1) also look inconsistent when comparing genetic controls in (Figure 6-Figure Supplement 1D) or when comparing the dilp mutants. Plotting this data as total sleep amount in the day & night (2 separate graphs) would be beneficial. It would also be helpful to see additional sleep traces for these experiments.

According to the reviewer's comments, we have added the sleep amounts of added *dilp3* and *dilp5* adults (Figure 6-figure supplement 1C-D) as well as IPC silencing (Figure6-figure supplement 3D) in a daytime/night time sleep-separated manner.

(19) For Figure 6, what effect does thermogenetic activation of hugin neurons have on IPC activity? The authors demonstrate in Figure 5 that thermal activation results in an increase in larval IPC activity, but they do not show these experiments in the adult brain. These experiments would provide more support for their conclusion that hugin has differential effects on IPC activity depending on the developmental age (larvae vs adults).

According to the reviewer's comments, we performed thermo-activation of hugin neurons and found no significant effects on adult IPCs (see Author response image 3), consists with the ex vivo data in Figure 6.



Author response image 3.

(20) A figure legend is needed for Figure 7. The model is not self-explanatory, nor is there an adequate explanation in the discussion section.

We have added legends (line 781-785).

(21) Since *hugin* is known to be downstream of *Dh44* in larvae, the discussion needs to include comparison to published work on *Dh44* in larvae (Poe et al, 2023). The *hugin* receptor, *PK2R1*, is also expressed in *Dh44* & *DMS* neurons (Schlegel et al, 2016), so a discussion of what role *Dh44*/*DMS* neurons may play in their model is necessary.

We agree. We have added discussion as follow in Discussion (line 313-320).

We cannot rule out the possibility that other neurons could function downstream of *Hugin*PC neurons in sleep regulation. For instance, given that *Dh44* neurons in the brain promote arousal (Poe et al. 2023) and are *PK2-R1*-positive (Schlegel et al. 2016), *Hugin* might control sleep in part through *Dh44* neurons.

(22) Minor point: Line 97 should say "resulted in a significant sleep increase." Currently, it says "decrease" which is not what is depicted in the figure.

We appreciate the reviewer's point. We have corrected this.

(23) Minor point: Figure 5 should be renamed as Figure 4 since the text describing the results in Figure 5A & 5B occurs before the text describing the results in Figure 4.

We do understand the point the reviewer arose. However, since Fig5A explains the experimental setup of the whole Fig5s, we would like to keep Fig5A at the original position.

Reviewer #2 (Recommendations for the authors):

First, the study would benefit from a more comprehensive discussion of previous research, particularly the work by Schlegel et al. (2016) and Melcher and Pankratz (2006). A key inconsistency that should be addressed is the observation that *hugin* mutant larvae exhibit reduced body size and feeding behavior, which may influence *Dilp2* secretion. The selective effect on *Dilp3* and *Dilp5* without affecting *Dilp2* warrants further clarification. Conducting conditional gene expression experiments to control *hugin*, *dilp3*, and *dilp5* expression, along with neuronal activity modulation, would help determine whether the observed effects are direct or secondary consequences.

According to the review's comments, we tried to manipulate neuronal activity in IPCs, but unfortunately, expression of *Kir2.1* in IPCs caused die or very weak animals. Instead, we cited a recent paper that shows a differential secretion of *Dilp2* and *Dilp6* in a stimulant-dependent

manner (Suzawa et al. PNAS 2025) and added more discussion about selective Dilp3/5 secretion by Hugin-PK2-R1 signals (line 275-297).

Second, the specificity of IPC secretion mechanisms should be clarified. Given that IPCs coexpress Dilp2, Dilp3, and Dilp5, it remains unclear how the pathway selectively modulates Dilp3 and Dilp5 while leaving Dilp2 unaffected. Additional experiments, such as electron microscopy, could provide insights into whether anatomical differences in vesicular pools influence peptide secretion. Since hugin mutants are reported to have reduced body size, confirming that Dilp2 secretion remains truly unchanged is crucial for eliminating potential indirect effects.

We thank this reviewer for the valuable suggestions. Since the selective Dilp secretion mechanisms in IPCs are not the main scope in this paper, we would like to attempt detailed EM analysis in next studies. We cited a recent paper that shows a differential secretion of Dilp2 and Dilp6 from IPCs in a stimulant-dependent manner (Suzawa et al. PNAS 2025) and added more discussion about selective Dilp3/5 secretion by Hugin-PK2-R1 signals (line 275-297).

Third, the study should explore the potential role of alternative circuits, such as the HuginPCDH44 pathway, in sleep regulation. The observation that DH44 mutants exhibit even greater sleep amounts than PK2-R1 mutants suggests the involvement of additional regulatory mechanisms. Prior studies indicate that HuginPC neurons may influence DH44 neuron activity, which could impact sleep. Furthermore, recent findings link DH44 with starvation-induced sleep loss in adult flies. Discussing and experimentally investigating the HuginPC-DH44 axis in larval sleep regulation would provide additional depth to the study.

As far as we understand, any direct evidence for HuginPC → DH44 pathway has not been reported in larvae as well as adults. Instead, DH44 influences Hugin neuron activity in adults (King et al. 2017). We thus examined whether optogenetic DH44 activation could influence HuginPC activity using CRTc analysis, but unfortunately, we could not detect significant changes in HuginPC activity.

Given that PK2-R1 is expressed in DH44-positive neurons (Schelgel et al 2016) and that DH44-positive neurons are localized at the regions to which HuginPC neurons innervate, it is still possible that the HuginPC → DH44 pathway might function in parallel to the HuginPC → IPCs pathway. We feel that this is quite an interesting possibility and should be a nice scope in the next paper.

Fourth, validating the functional connectivity between HuginPC neurons and IPCs using calcium imaging would significantly enhance the study. Employing real-time calcium imaging with GCaMPs would provide direct evidence of synaptic activity between these neuronal populations. Such data would strengthen the claim that the observed sleep regulatory effects result from direct neural communication rather than secondary systemic influences.

We agree. Indeed, we tried Ca²⁺ imaging of HuginPC neurons and IPCs in living larvae as well as using ex vivo preparations, and realized that it was quite technically difficult to obtain reliable Ca²⁺ dynamics data in the brain of living larvae/ex vivo brain tissue. Therefore, instead of live Ca²⁺ imaging, we performed the CRTc analysis using fixed brain preparations. We have added the mention that we tried Ca²⁺ imaging in the larval brain, but it did not work well (line 555-558).

Finally, a more detailed discussion of developmental differences in sleep regulatory mechanisms would be beneficial. The manuscript should address why genes involved in sleep modulation during development may function differently from their roles in adult sleep regulation. Providing a conceptual framework or experimental evidence to explain these developmental differences would enhance the study's contribution to

understanding the evolution of sleep circuits. Clarifying how these findings fit into broader sleep regulation models would increase the impact of the research.

We agree. We would like to add discussions about how factors/circuits involved in sleep modulation during development may function differently from their roles in adult sleep regulation as follows (line 349-371), as it is rather difficult to discuss why.

It is thus possible that Hugin/PK2-R1 signaling along the HugPC-IPCs circuitry is suppressed in adults. IPCs in adults receive multiple positive and negative modulatory inputs through GPCRs including the metabotropic GABA_B receptors (Enell et al., 2010), which suppresses IPC activity and Dilp release in adult IPCs (Enell et al., 2010). It is thus plausible that such negative modulatory inputs to IPCs in adults might counteract with the Hugin/PK2-R1 axis to suppress Dilp release. In addition, our data suggest that Dilps modulate sleep amount in the opposite directions in larvae and adults (Figure 7). Comparing the expression levels and activities of GPCRs in larval and adult IPCs would be essential to better understand how the same modulatory signals over the course of development come to exert differential impacts on sleep. Interestingly, Hugin in adults appears irrelevant for the baseline sleep amount but is required for homeostatic regulation of sleep (Schwarz et al., 2021). Thus, testing if Hugin/PK2-R1 axis is involved in the homeostatic regulation of larval sleep, and how such a system compares to its adult counterpart, may further provide mechanistic insights into how homeostatic sleep regulation matures over development.

By addressing these aspects, the manuscript will provide a clearer, more robust, and well-supported analysis of larval sleep regulation. These refinements will help improve the study's clarity and impact, ensuring that its findings are effectively communicated to the research community.

Reviewer #3 (Recommendations for the authors):

(1) Line 97: "Silencing neurons expressing Oamb and PK2-R1 resulted in a significant sleep decrease?" But there is an increase in sleep amounts from Figure 1A. (Typo error).

We thank the reviewer for pointing out this typo. We have corrected this typo in the revised version.

(2) Line 139: "HugPC and IPCs labeled by Dilp3-GAL4 are located in close proximity to each other." While proximity does not equal synaptic connections, direct connectivity of HugPC and IPCs was already shown in larval connectome analyses with HugPC providing the strongest input of larval IPCs (Hückesfeld et al. eLife 2021). This could be cited in this context instead.

We agree. We have cited this paper in References (line 163).

(3) Figure 2 Supplement 1: Locomotion speed is affected in PK2-R1 knockouts; what is the significance regarding the observed sleep increase?

We agree that this is a very important point. As the reviewer pointed out, since locomotion speed was reduced in PK2-R1 KO larvae, sleep increase phenotype in PK2-R1 KO larvae might be in part due to reduction of locomotion. On the other hand, IPCs silencing by Kir2.1 caused sleep increase phenotype without significant changes in locomotion (Figure 2; Figure 2-figure supplement 1). It is thus possible that since PK2-R1 is broadly expressed in the nervous system in addition to IPCs (Figure 2), PK2-R1 neurons other than IPCs might contribute to locomotion control.

(4) Why are Dilp3 levels changing (increasing) in adult IPCs after PK-2 treatment? This is not mentioned in the text and is not discussed at all.

As the reviewer indicated, this data is unexpected to us. At this moment, we could only assume that PK-2 could act in larval and adult IPCs in a different manner. We have added this sentence in Results (line 211-214).

(5) *It has been shown in other publications that Dilps play a role in sleep regulation (Cong et al., Sleep 2015), this study should be cited.*

We have cited this paper in References (line 224).

(6) *The order of discussing figure panels is sometimes confusing, e.g. Figure 6C is discussed at the very end after 6D-F.*

We agree. Indeed, we discussed a lot about this order during preparation of the first draft. However, we finally decided the current order, as grouping “sleep phenotype data” and “ex vivo data” should be easier to understand for readers. We thus keep the current order in the revised submission.

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